



DUEL COUNTER BALANCE VALVE Model : 2CB06T03

Ref. No. D 28151
Release: 08 / 2026

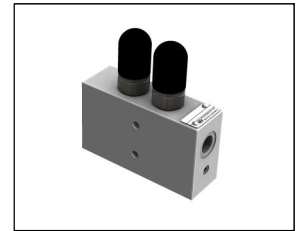
ENGINEERING - 1 of 2

A Polyhydon Group Company

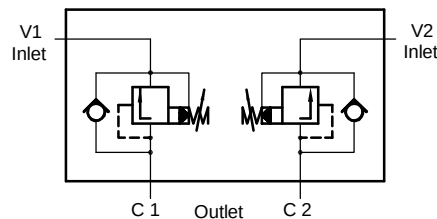
Description

Duel counter balance valves 2CB06T03 are spool type valves. They offer free flow from their port **V1** to Port **C1** and creates a back pressure or a load in opposite direction upto a predetermined pressure setting. The predetermined pressure setting can be adjusted within its maximum specified range.

This valve is made for a special purpose application. Please consult Polyhydon before placing the order.



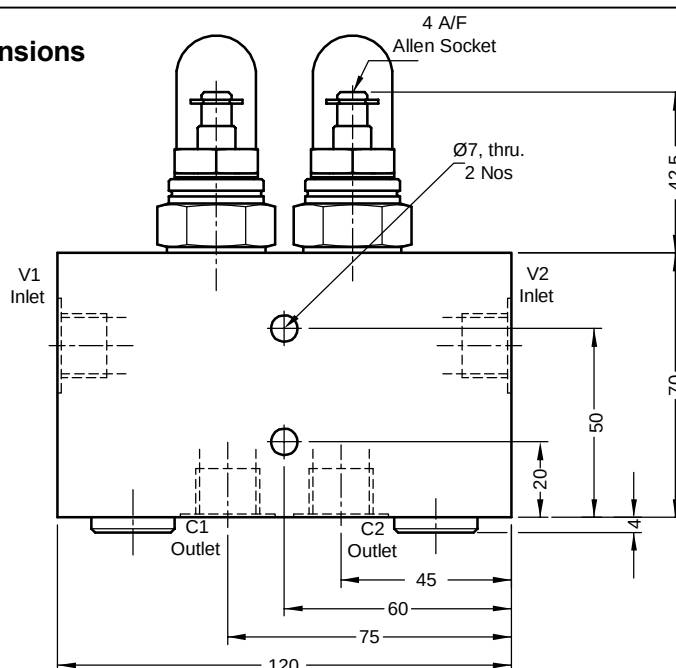
Hydraulic symbol



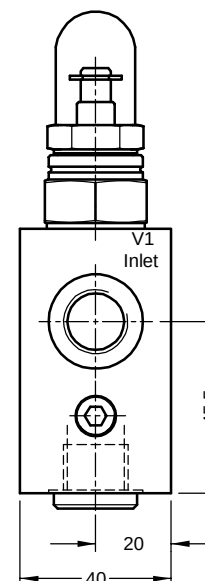
Technical specifications

Construction	:	Spool type, Internally pilot operated.
Mounting style	:	Threaded port body
Inlet and outlet ports size	:	G $\frac{3}{8}$.
Flow direction	:	Free flow from V1 to C1
	:	Piloted flow in opposite direction.
Check valve Cracking pressure	:	1 bar.
Working pressure	:	315 bar for all Ports.
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Max. flow handling capacity	:	30 l/min.
Mass approx.	:	2.42 Kg.

Unit dimensions



Dimensions in mm.



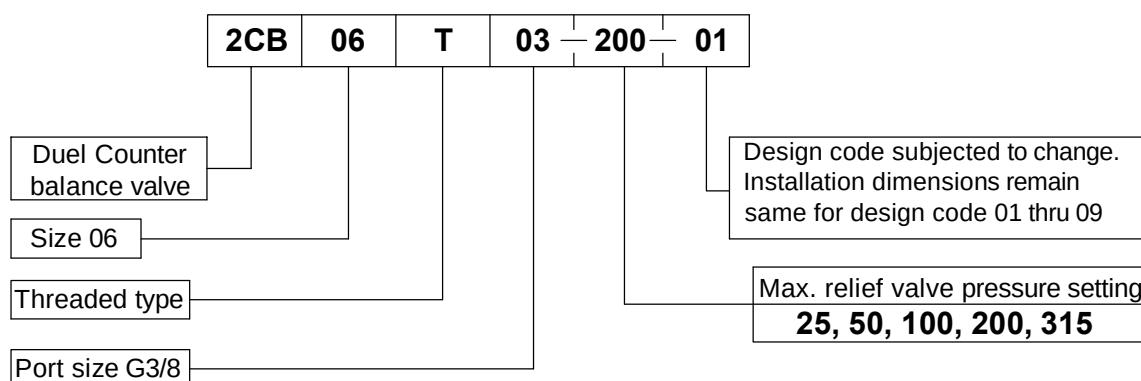
polyhydon pvt. ltd.

78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
E-mail : sales@polyhydon.com
Website : www.polyhydon.com



Ordering code





COUNTER BALANCE VALVE Model : CBS20 (30 Series)

Ref. No. D 03449
Release: 01 / 2026

ENGINEERING - 1 of 3

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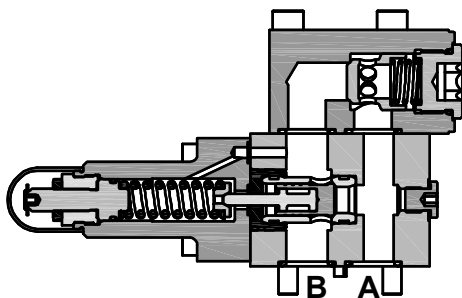
Description

Counter balance valves CBS20*** are seat type valves. These offer free flow from port **B** to Port **A** and give leak free closure in opposite direction upto a predetermined cracking pressure. The predetermined cracking pressure can be adjusted within its maximum specified range. For higher cracking pressure or external leakage drain, Port **X** can be used.

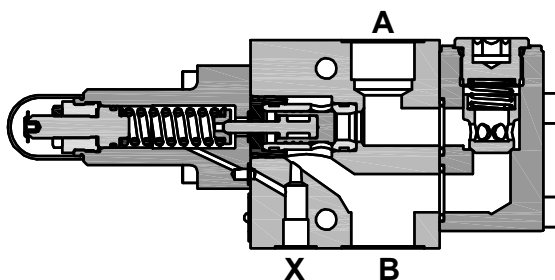


Section

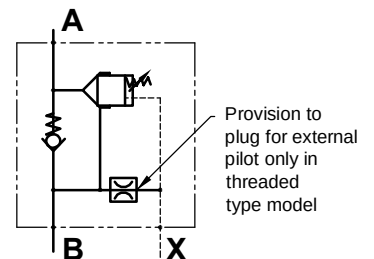
Subplate type



Threaded type



Hydraulic symbol

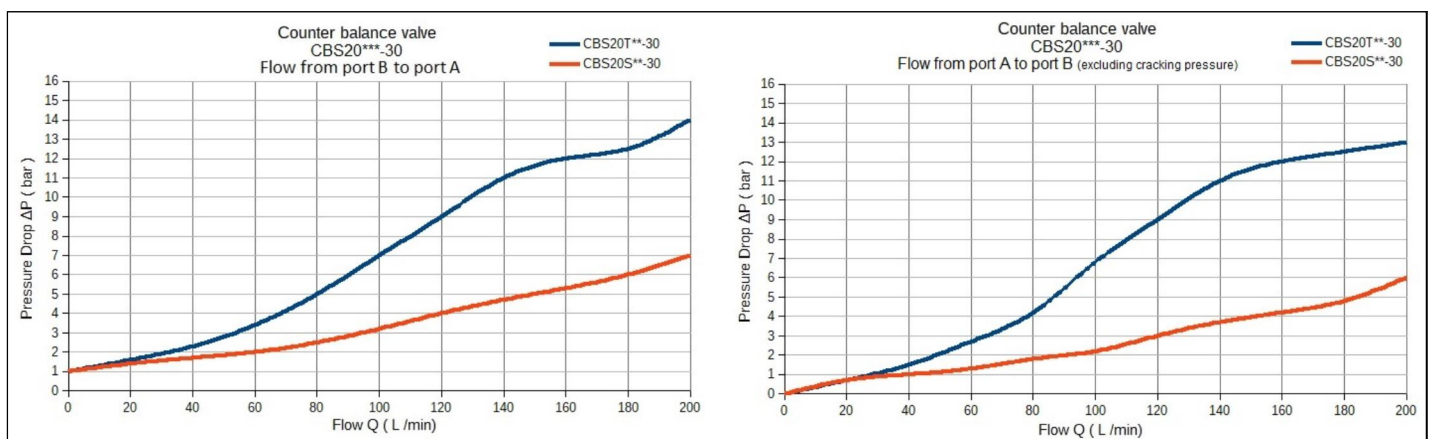


Technical specifications

Construction	:	Direct acting , Seat type.
Mounting style	:	Threaded port body and subplate body
Flow direction	:	Free flow from B to A
	:	Counter balance function from port A to port B.
Cracking pressure of counter balance function	:	Refer ordering code.
Working pressure	:	315 bar for all Ports.
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Max. flow handling capacity	:	200 l/min max.
Mass approx.	:	Model CBS20T CBS20S
	in Kg.	8.6 6.9

Performance graph

Oil grade used VG46 at 30°C



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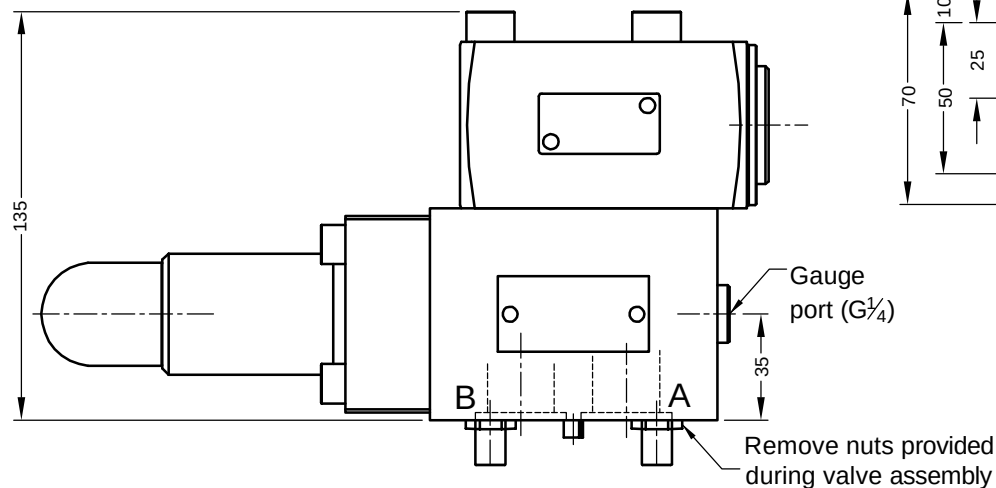
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Website : www.polyhydron.com



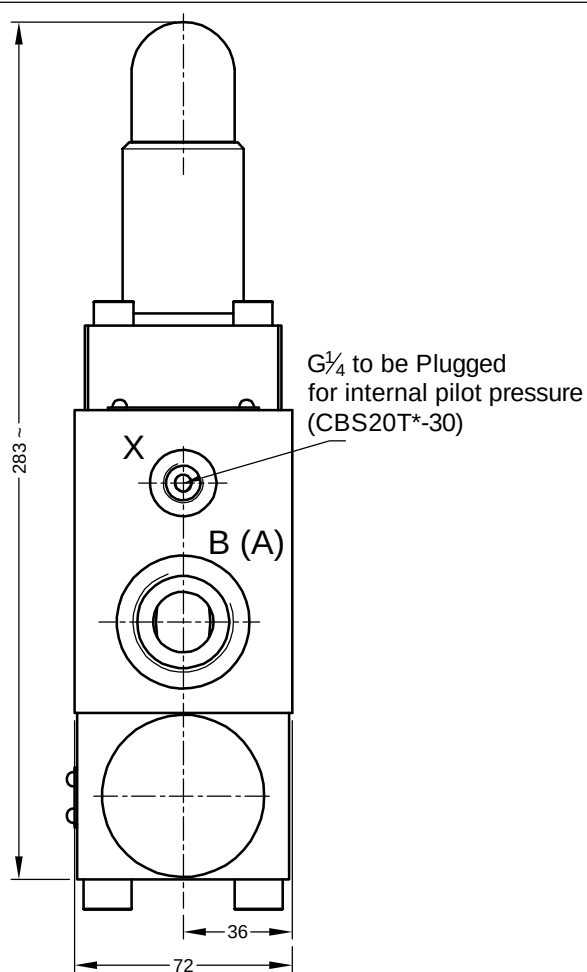
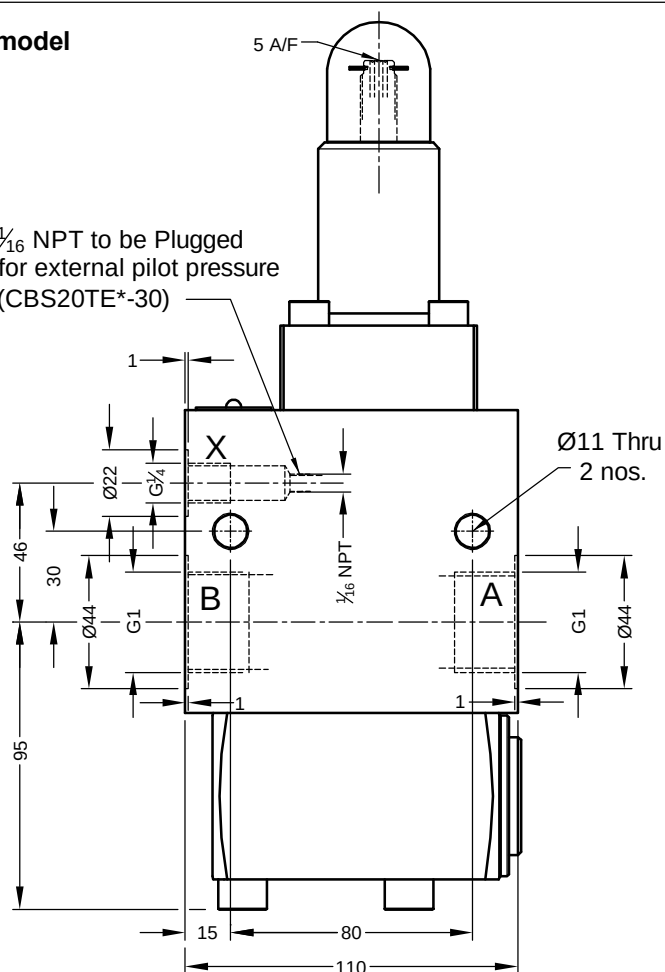
ENGINEERING - 2 of 3

Subplate model



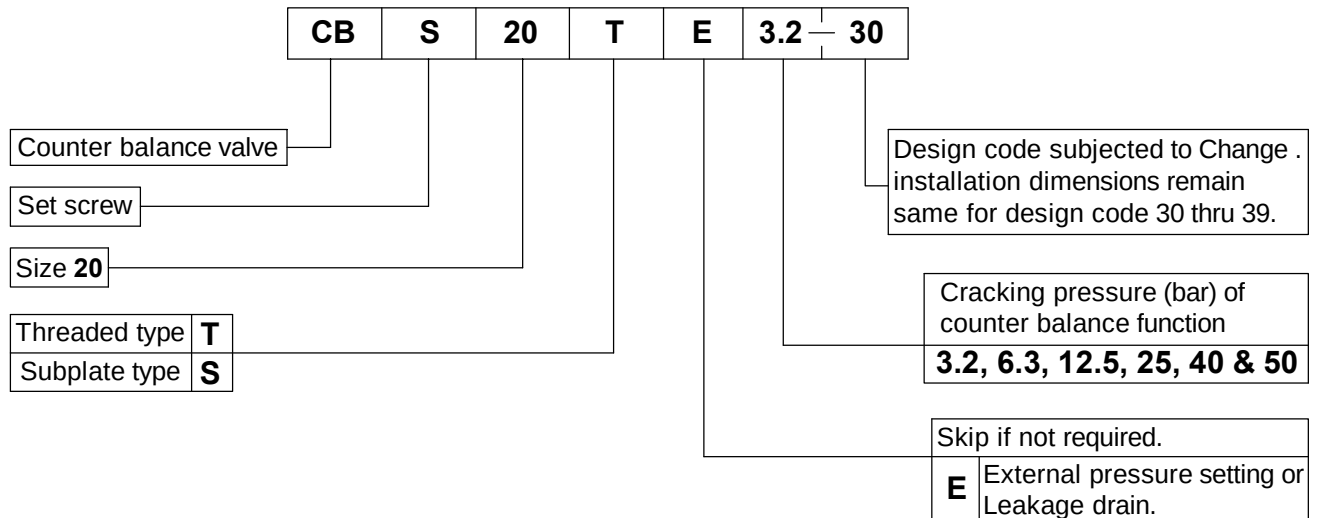
Technical drawing of the front view of a square plate. The overall dimensions are 75 mm by 70 mm. The plate features two large circular openings, labeled A and B, with a maximum diameter of Ø22 mm. There are four smaller circular openings, labeled X, with a diameter of Ø6 mm. The plate is made of M10x1.5 18dp, 4nos. material. The drawing includes various dimension lines and callouts: 75 (total width), 55 (distance from right edge to center of opening A), 45 (distance from left edge to center of opening B), 27.5 (distance between centers of openings A and B), 10 (distance from top edge to center of opening A), 10 (distance from top edge to center of opening B), 25 (distance from bottom edge to center of opening A), 50 (distance from bottom edge to center of opening B), and 10 (distance from center of opening A to center of opening X).

1/16 NPT to be Plugged
for external pilot pressure
(CBS20TE*-30) —





Ordering code



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Due to continuous improvement in the design of the product, the actual product supplied may look different than shown above.
For critical applications, please ask for certified installation drawing.



COUNTER BALANCE VALVE

Model : CBS***T***

Ref. No. D 02241
Release: 01 / 2026

ENGINEERING - 1 of 2

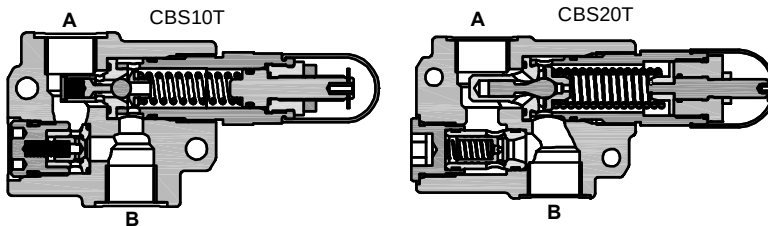
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Description

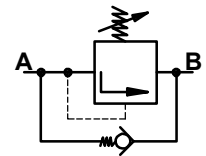
Counter balance valves CBS***T*** are seat type valves. They offer free flow from their port **B** to Port **A** and give leak free closure in opposite direction upto a predetermined cracking pressure. The predetermined cracking pressure can be adjusted within its maximum specified range.



Section



Hydraulic symbol

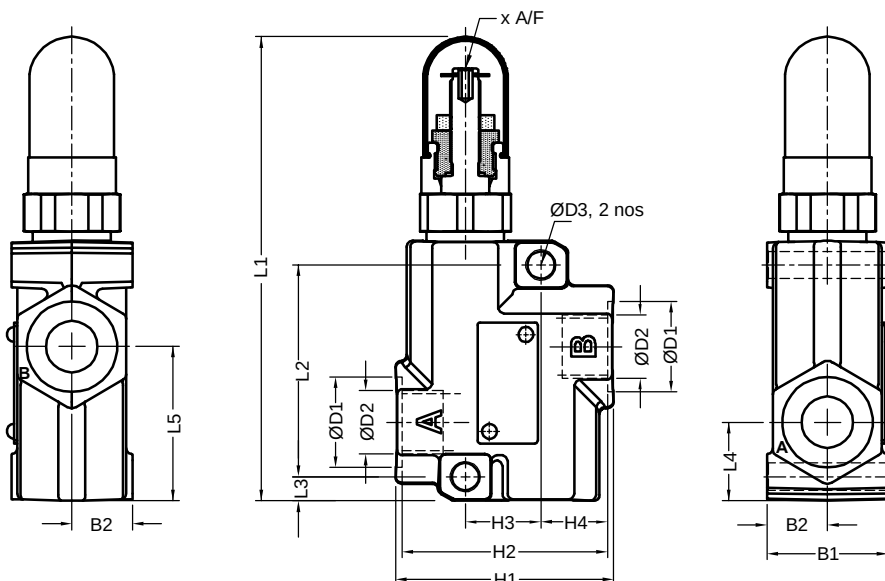


Technical specifications

Construction	:	Direct acting , Seat type, Internally pilot operated.		
Mounting style	:	Threaded port body		
Flow direction	:	Free flow from B to A		
	:	Piloted flow in opposite direction.		
Cracking pressure	:	Refer ordering code.		
Working pressure	:	315 bar for all Ports.		
Hydraulic medium	:	Mineral oil.		
Temperature range	:	-20°C to + 80°C.		
Viscosity range	:	10 cSt to 380 cSt.		
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.		
Max. flow handling capacity	:	Model	CBS10T	CBS20T
		in l/min.	30	115
Mass approx.	:	Model	CBS10T	CBS20T
		in Kg.	1.3	4.0

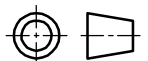
Unit dimensions

Dimensions in mm.



Size	10	20
L1	154	226
L2	70	110
L3	8	11
L4	26	41
L5	51	86
B1	40	60
B2	20	30
H1	72	100
H2	68	96
H3	25	50
H4	22	23
ØD1	G1/2	G1
ØD2	30	44
ØD3	9	11
x A/F	4	5

Note : Valve mounting screws are out of scope of supply.



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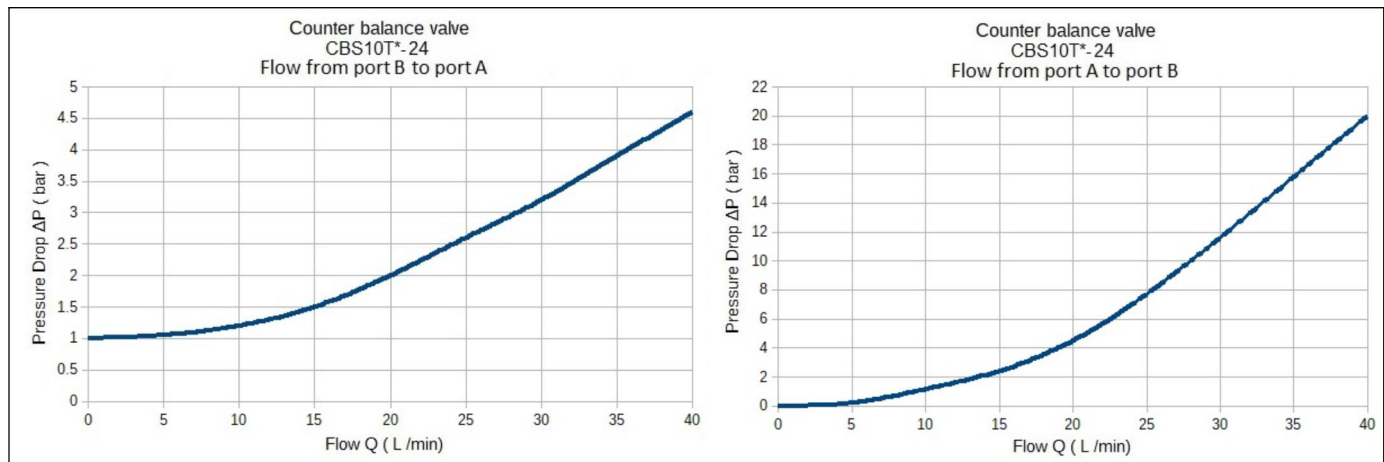
Phone : +91-(0)831- 2411001
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Website : www.polyhydon.com



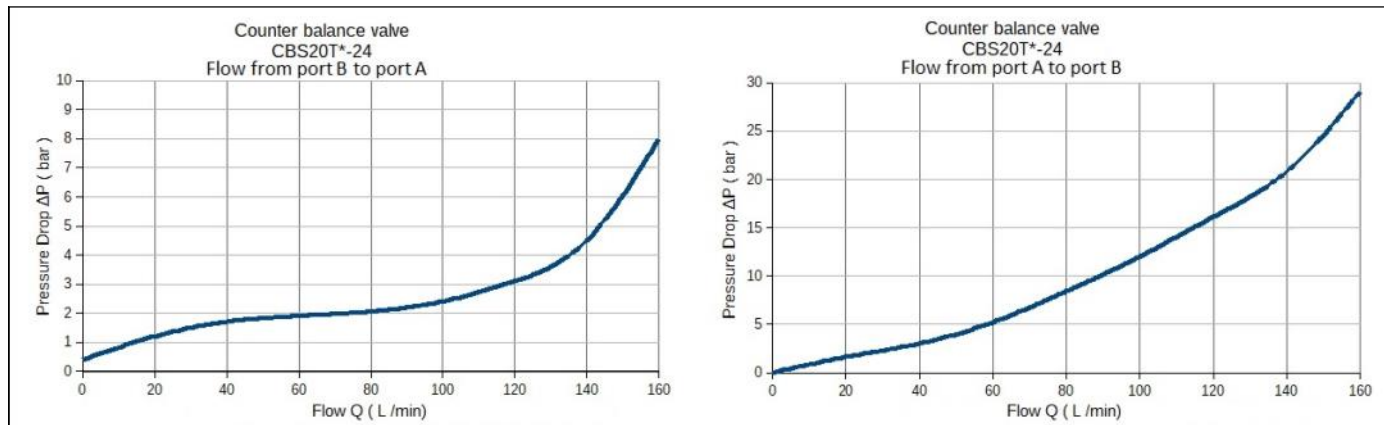
Performance graph

Grade of Oil used VG46 at 30°C temperature.

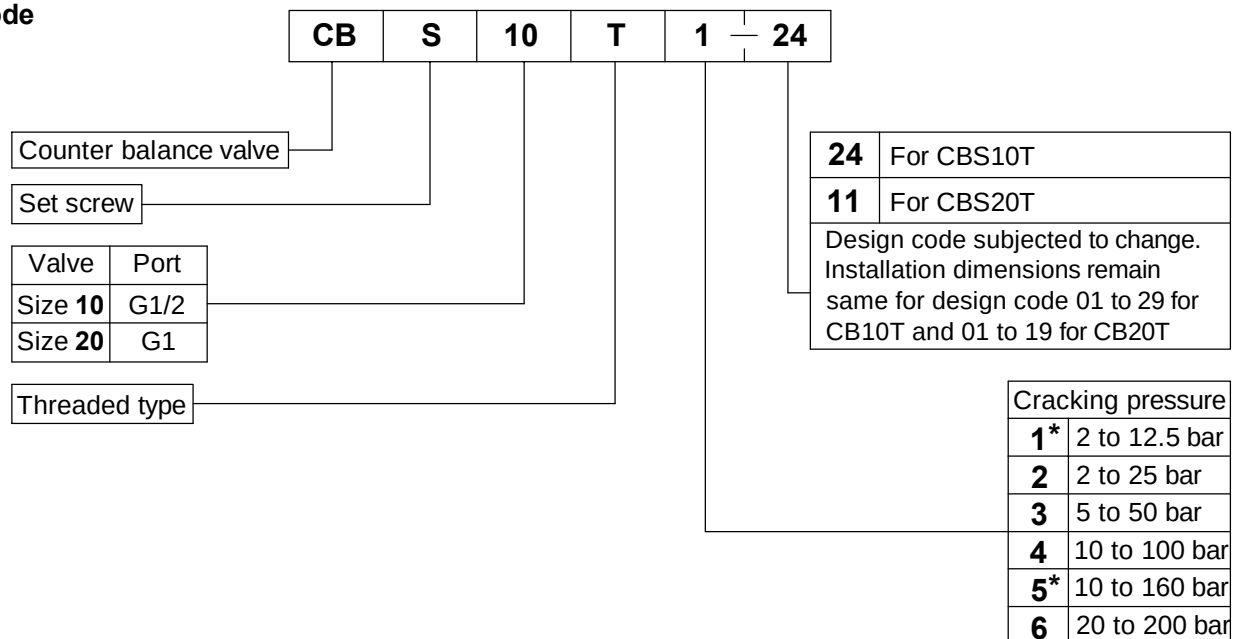
CBS10T*-24



CBS20T*-24



Ordering code



Note : Cracking pressure marked with *
are not available in valve size 20.

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For critical applications, please ask for certified installation drawing.



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PRESSURE CONTROL MODULE PCM 30-16 Model : PCM3016*****

Ref. No. D 04399
Release 12 / 2025
ENGINEERING - 1 of 3

Description

Design to control double pumps of Hi-Low system.

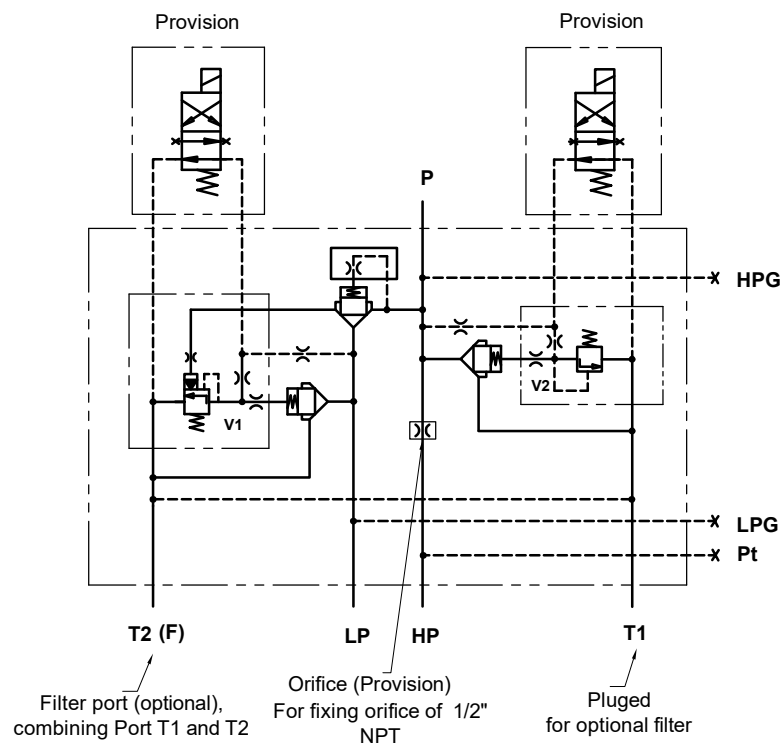
Unloads low pressure pump, when system pressure rises above the pressure set on Unloader valve.

Relieves high pressure pump, when system pressure reaches Relief valve set pressure.

Solenoid unloading facility available as an option.



Hydraulic symbol :



Connection Details :

HP	:	Inlet Port for high pressure pump
LP	:	Inlet Port for low pressure pump
P	:	Out-let port
HPG	:	Gauge port on high pressure pump line
LPG	:	Gauge port on low pressure pump line
Pt	:	Pilot pressure port (optional)
T1	:	Tank port of high Pressure Relief valve
T2	:	Tank port of low Pressure Unloading valve
F	:	Filter port connection optional (Combining T1 & T2)

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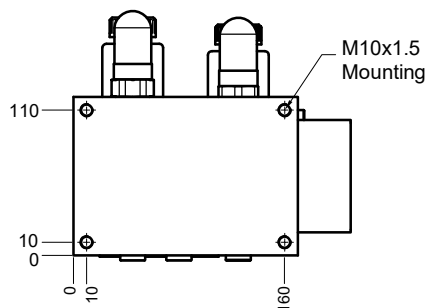
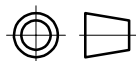
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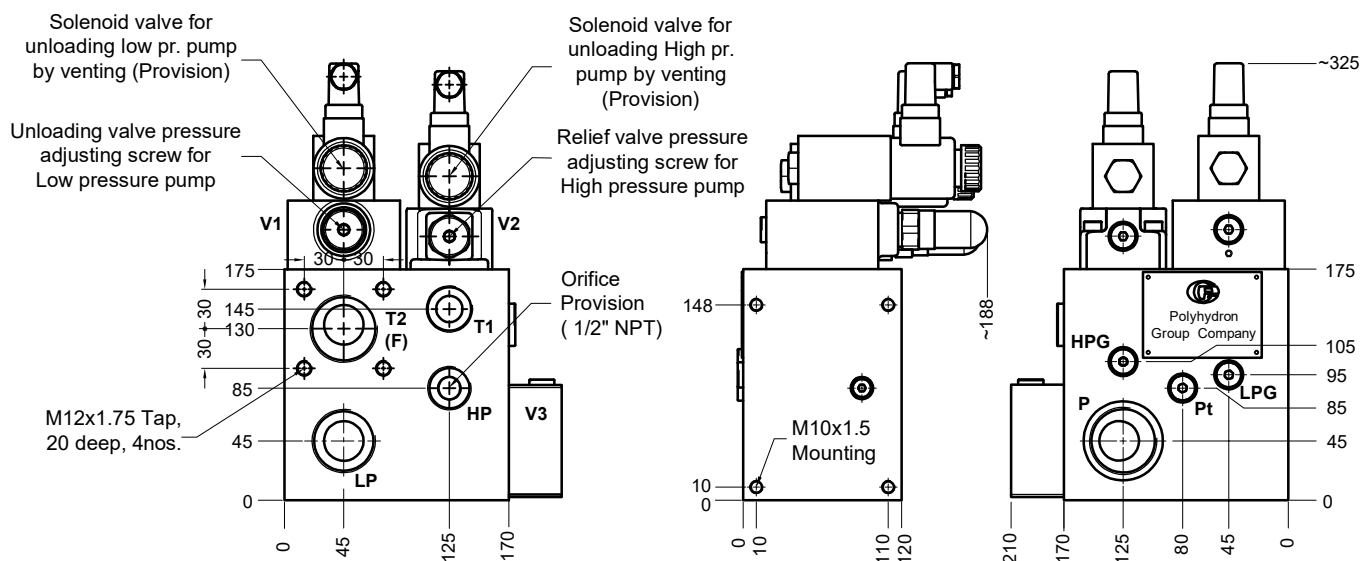


Unit dimensions

Dimensions in mm.



Port Details	
P	G 1 1/2
T1	G 1
T2 (F)	G 1 1/2
LP	G 1 1/2
HP	G 1
LPG	G 1/4
HPG	G 1/4
Pt	G 1/4

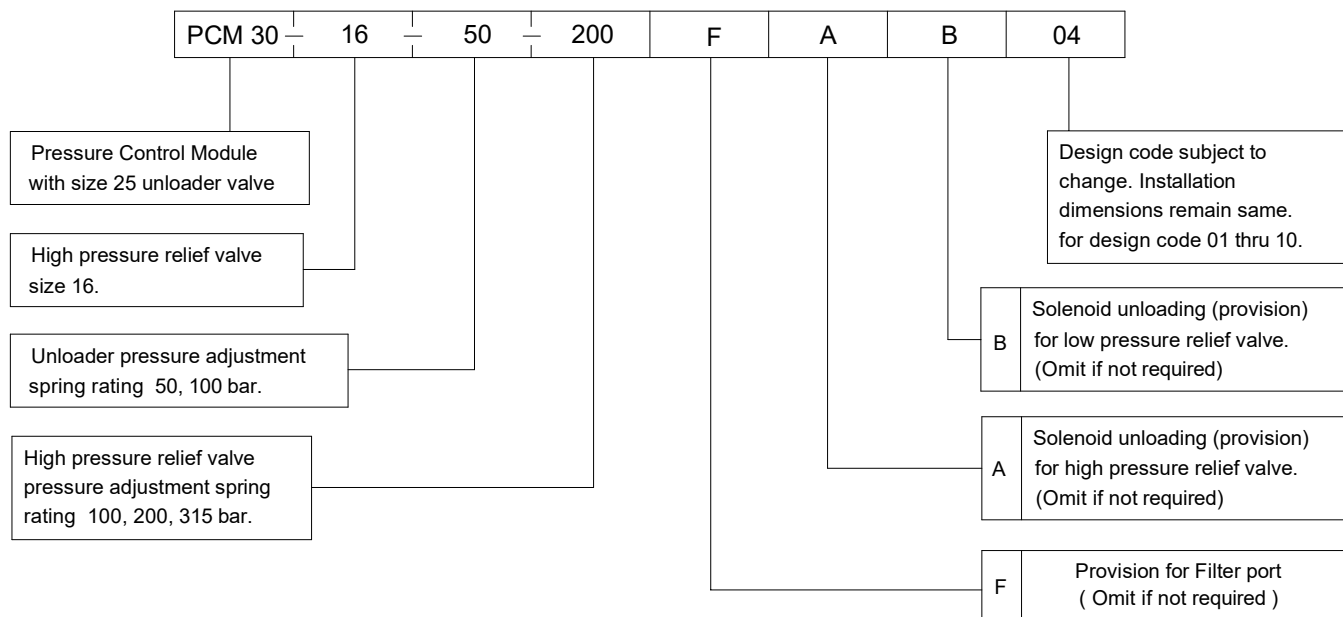




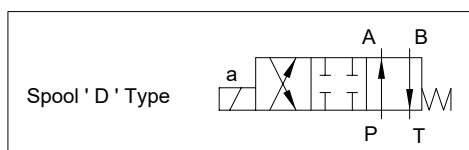
Technical specifications

Construction	:	Two stage poppet type high pressure Relief valve Two stage poppet type low pressure unloading valve
Mounting / Standard	:	Threaded body (Factory standard).
Mounting position	:	Optional.
Flow direction	:	As per Hydraulic Symbol.
Maximum operating pressure	:	For Low pressure pump 100 bar. For high pressure pump port 315 bar For port 'P' 315 bar.
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	High Pressure pump maximum flow : 100 lpm. Low Pressure pump maximum flow : 400 lpm
Mass (approx)	:	28 kg Approx.

Ordering code



Note : Solenoid (refer spool type) and Subplate to be ordered separately.





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PRESSURE CONTROL MODULE **PCM 20-16** **Model : PCM20-16*******

Ref. No. D 04902
Release 01 / 2026
ENGINEERING - 1 of 3

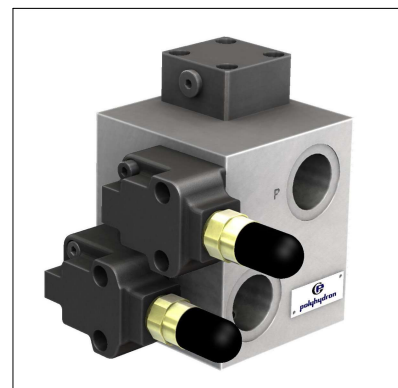
Description :

Design to control double pumps of Hi-Low system.

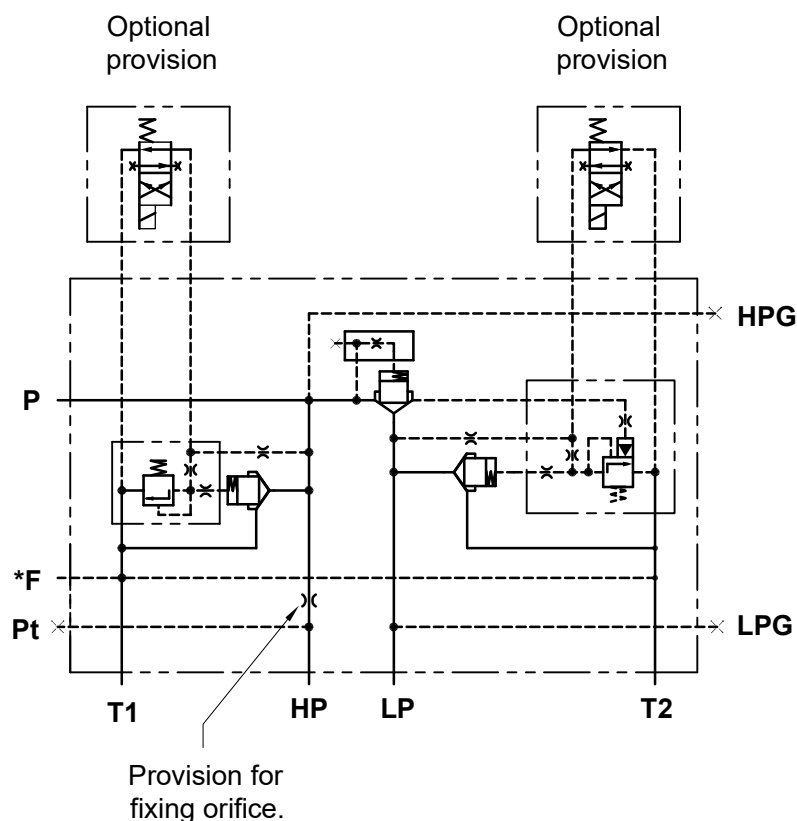
Unloads low pressure pump, when system pressure rises above the pressure set on Unloader valve.

Relieves high pressure pump, when system pressure reaches Relief valve set pressure.

Solenoid unloading facility available as an option.



Hydraulic symbol :



Connection Details :

HP	:	Inlet Port for high pressure pump
LP	:	Inlet Port for low pressure pump
P	:	Out-let port
HPG	:	Gauge port on high pressure pump line
LPG	:	Gauge port on low pressure pump line
Pt	:	Pilot pressure port (optional)
T1	:	Tank port of high Pressure Relief valve
T2	:	Tank port of low Pressure Unloading valve
F	:	Filter port connection optional (Combining T1 & T2)

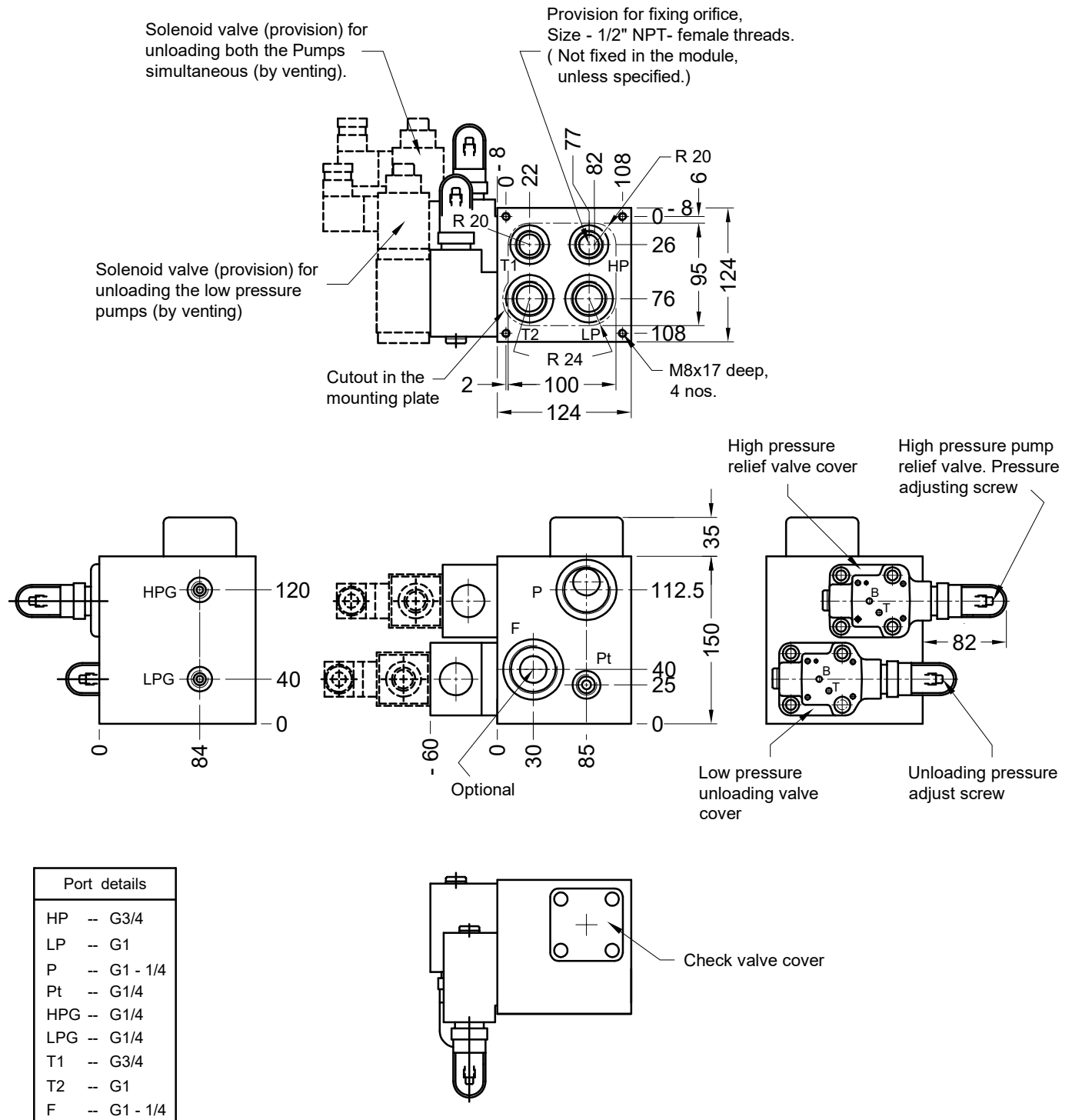
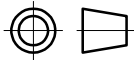
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78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com

Unit dimensions :

Dimensions in mm.



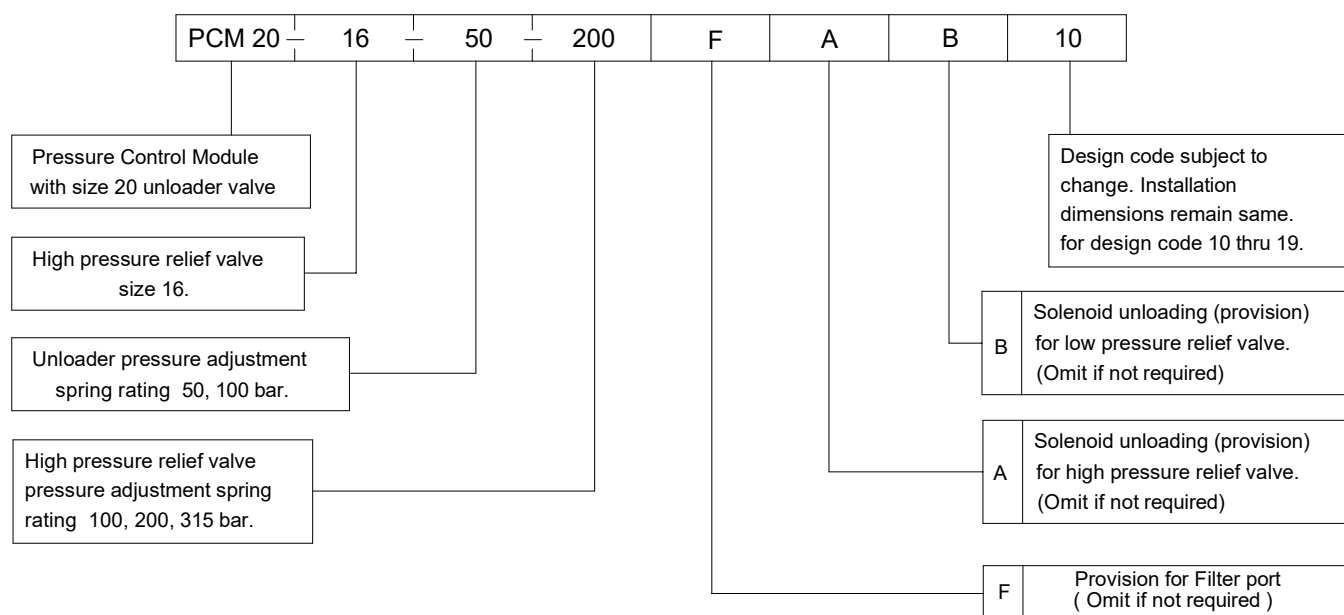
Note : Optional Solenoid to be ordered seperately.

**Technical specifications :**

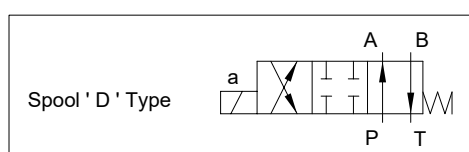
Construction PCM 20-16	:	Direct acting poppet type High Pr. Relief Valve
	:	Two stage poppet type low pressure unloading valve.
Mounting	:	Threaded body (Factory standard).
Mounting position	:	Optional.
Flow direction	:	As per Hydraulic Symbol.
Operating pressure	:	For Low pressure pump50 and 100 bar.
	:	For high pressure pump100, 200 and 315 bar.
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	

Model	High Pressure Max.	Low Pressure Max.
PCM 20-16	100 l/min	160 l/min

Mass : Approx 10.5 kg.

Ordering code :

Note : Solenoid (refer spool type) and Subplate to be ordered separately.





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PRESSURE CONTROL MODULE PCM 20-06 and PCM 20-10 Model : PCM20*****

Ref. No. D 06793
Release 01 / 2026
ENGINEERING - 1 of 4

Description :

Design to control double pumps of Hi-Low system.

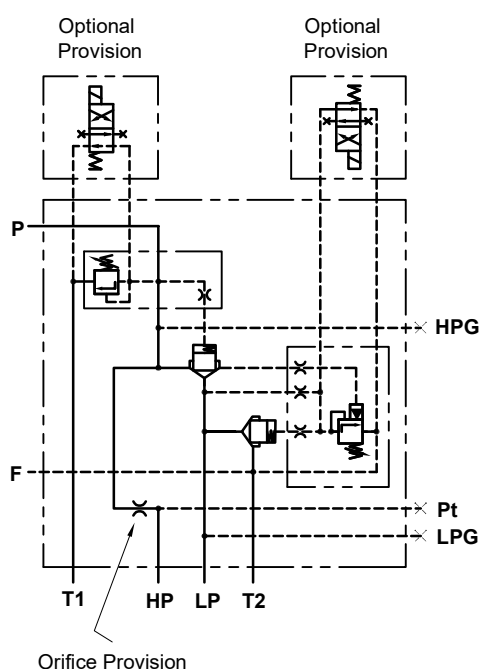
Unloads low pressure pump, when system pressure rises above the pressure set on Unloader valve.

Relieves high pressure pump, when system pressure reaches Relief valve set pressure.

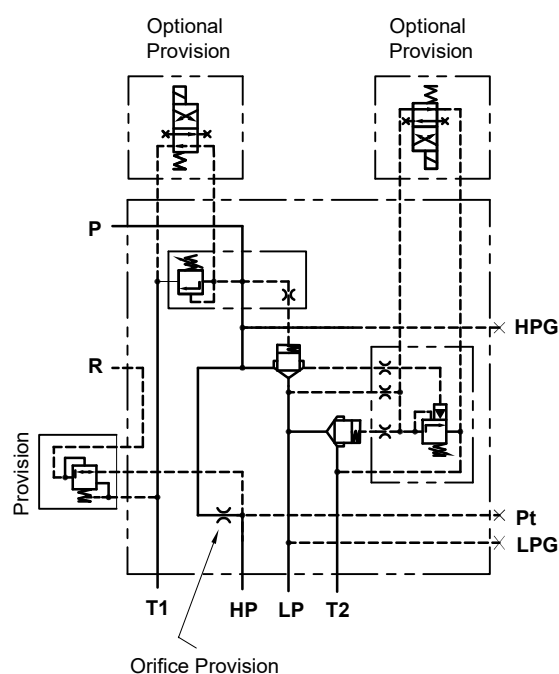
Solenoid unloading facility available as an option.



Hydraulic symbol :



PCM 20-06 and 20-10 with
filter port, relief valve tank line
combining T1 and T2 port.



PCM 20-06 and 20-10 with provision for
mounting Pressure reducing valve
DPM*06*A-01-*

Connection Details :

HP	:	Inlet Port for high pressure pump
LP	:	Inlet Port for low pressure pump
P	:	Out-let port
HPG	:	Gauge port on high pressure pump line
LPG	:	Gauge port on low pressure pump line
Pt	:	Pilot pressure port (optional)
T1	:	Tank port of high Pressure Relief valve
T2	:	Tank port of low Pressure Unloading valve
F	:	Filter port connection optional (Combining T1 & T2)
R	:	Outlet Port (Reduced Pressure)

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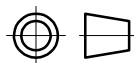
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Website : www.polyhydon.com



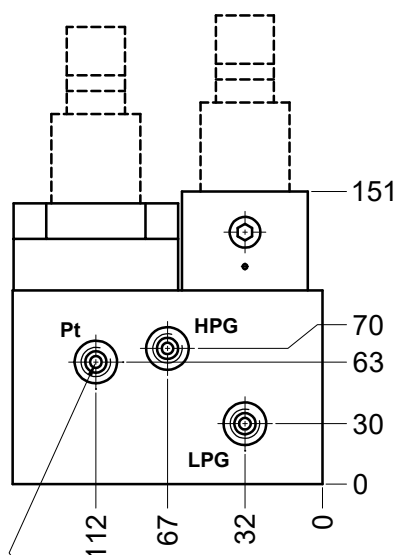
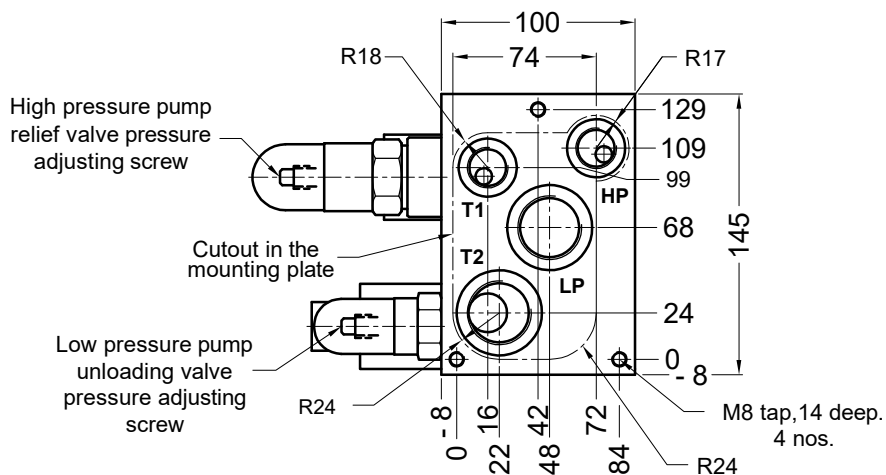
Unit dimensions :

Dimensions in mm.

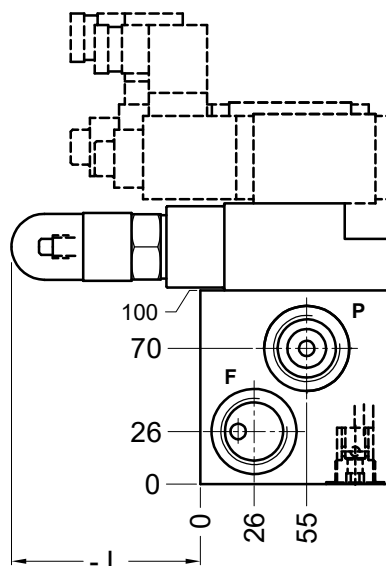


PCM 20-06 and PCM 20-10 with provision for filter port.

Port details	
HP	-- G1/2
LP	-- G1
P	-- G1
Pt	-- G1/4
HPG	-- G1/4
LPG	-- G1/4
T1	-- G1/2
T2	-- G1
F	-- G1



Open this plug to insert orifice in high pr. line (not fixed in the module unless specified) orifice size 1/8" NPT



Model	- L
PCM20/06	83
PCM20/10	98

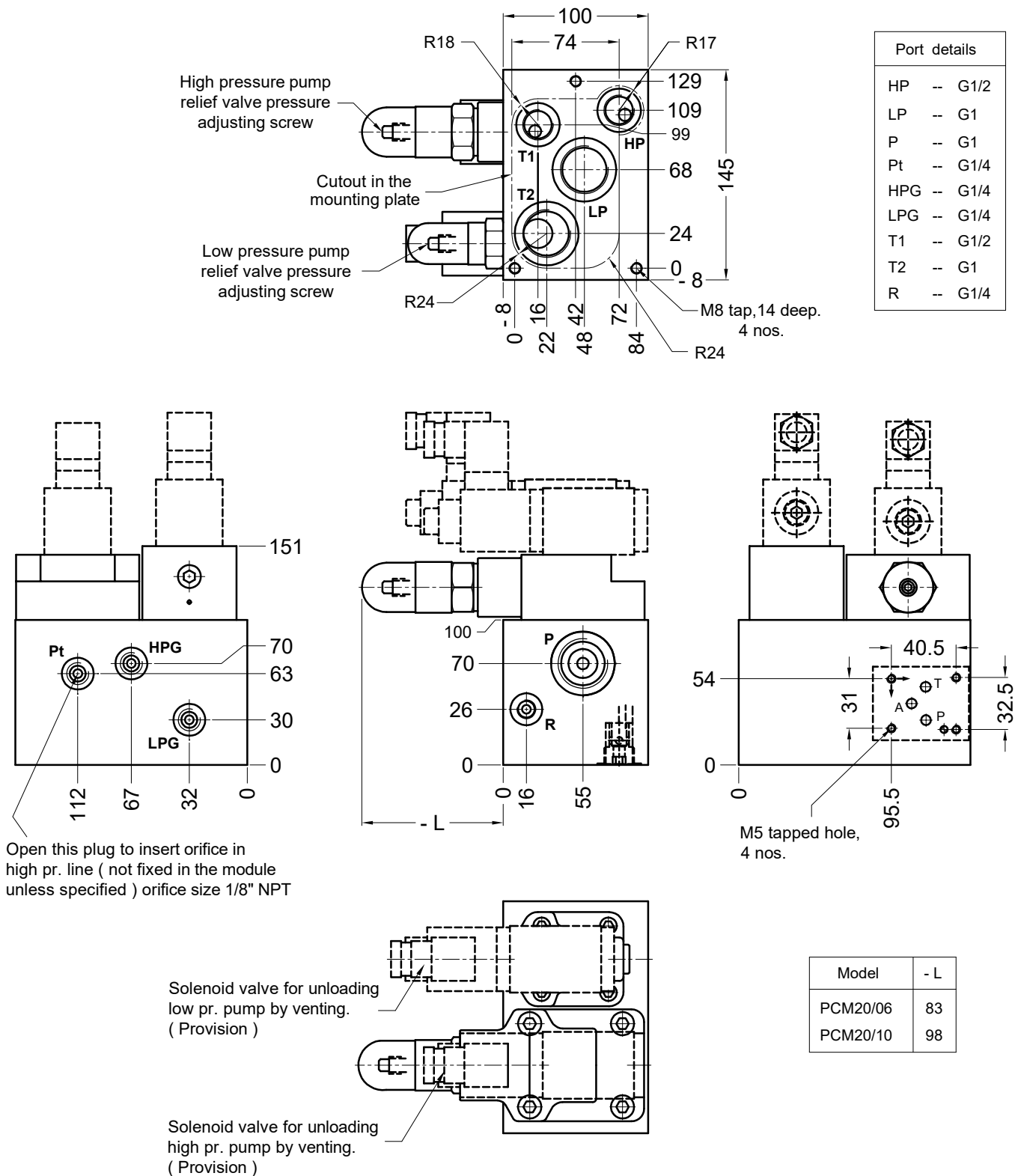
Solenoid valve for unloading low pr. pump by venting. (Provision)

Solenoid valve for unloading high pr. pump by venting. (Provision)

Note : Optional Solenoid to be ordered separately.



PCM 20/06 and PCM 20/10 with provision for mounting pressure reducing valve DPM*06*A-01*



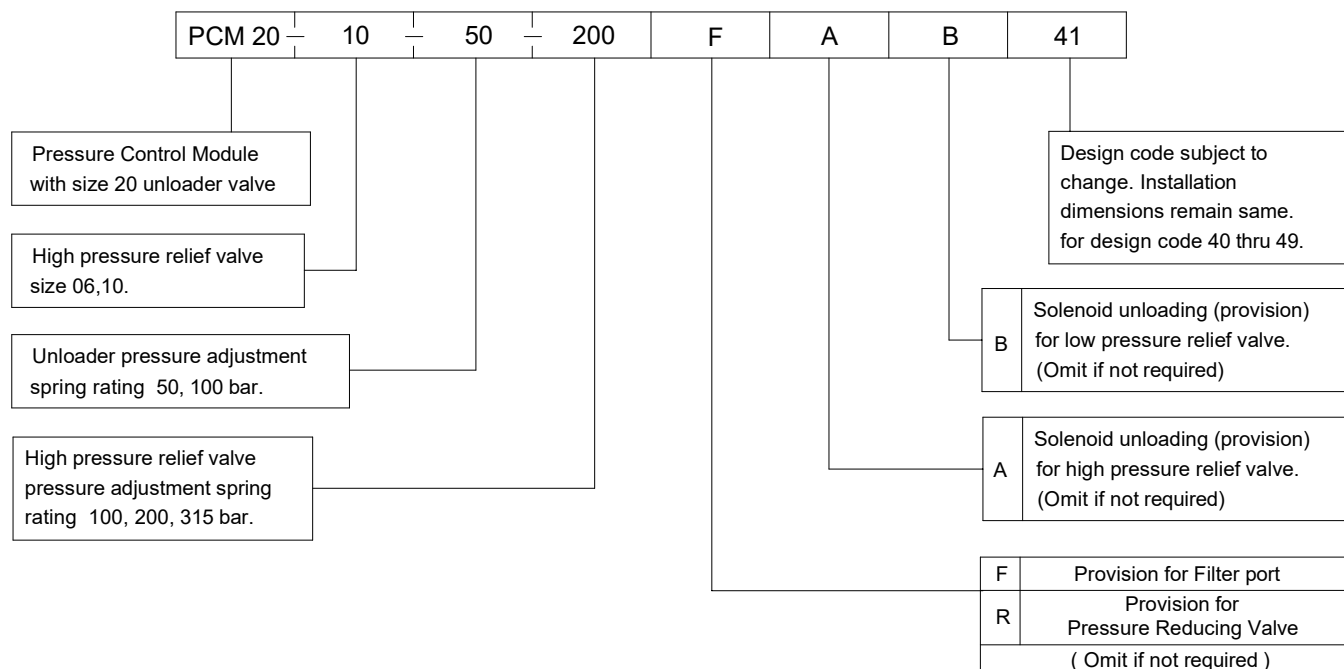
Note : Optional Solenoid to be ordered separately.

**Technical specifications :**

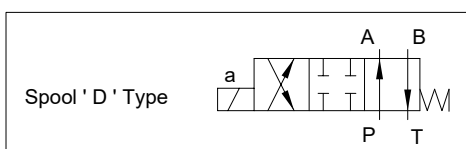
Construction PCM 20-06 and 20-10	:	Direct acting poppet type High Pr. Relief Valve Two stage poppet type low pressure unloading valve.
Mounting	:	Threaded body (Factory standard).
Mounting position	:	Optional.
Flow direction	:	As per Hydraulic Symbol.
Operating pressure	:	For Low pressure pump50 and 100 bar. For high pressure pump100, 200 and 315 bar.
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	

Model	High Pressure Max.	Low Pressure Max.
PCM 20-06	25 l/min	160 l/min
PCM 20-10	60 l/min	160 l/min

Mass : Approx 7.5 kg.

Ordering code :

Note : Solenoid (refer spool type) and Subplate to be ordered separately.





PRESSURE CONTROL MODULE **PCM 06-06** **Model : PCM06-06*******

Ref. No. D 03645
 Release 12/2025
 ENGINEERING - 1 of 2

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Description

Design to control double pumps of Hi-Low system.
 Unloads low pressure pump, when system pressure rises above the pressure set on Unloader valve.
 Relieves high pressure pump, when system pressure reaches Relief valve set pressure.

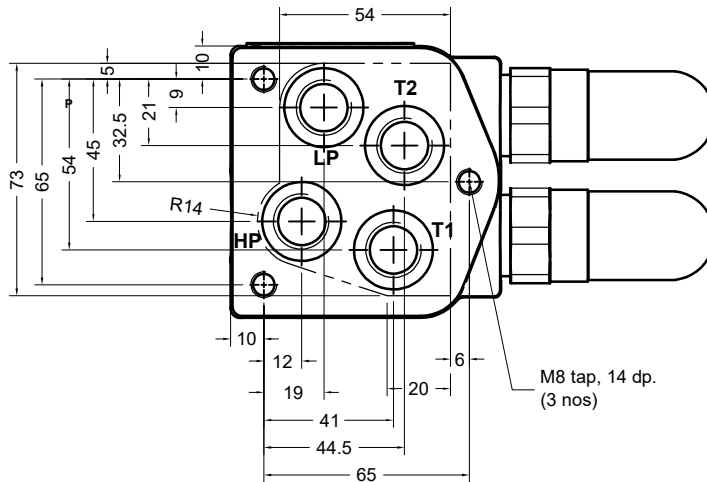
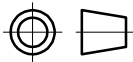
Connection Details :

HP : Inlet Port for high pressure pump
 LP : Inlet Port for low pressure pump
 P : Out-let port
 HPG : Gauge port on high pressure pump line
 LPG : Gauge port on low pressure pump line
 T1 : Tank port of high Pressure Relief valve
 T2 : Tank port of low Pressure Unloading valve

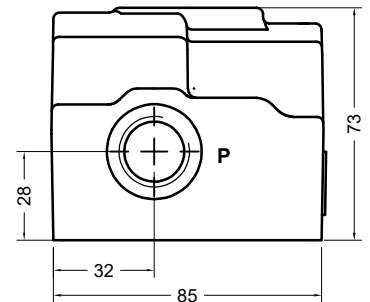
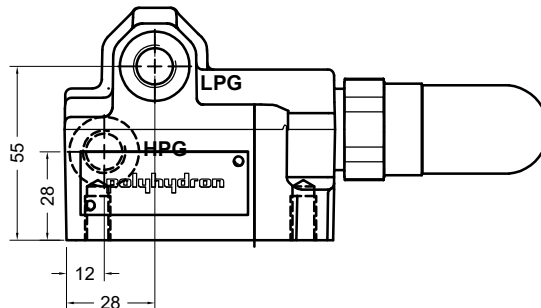
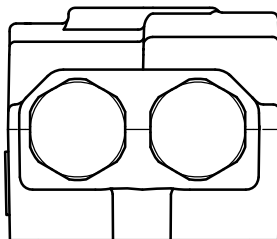
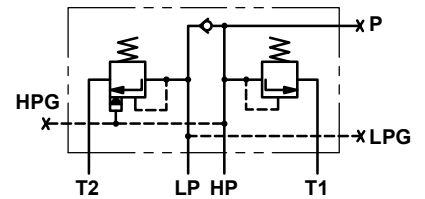


Unit dimensions

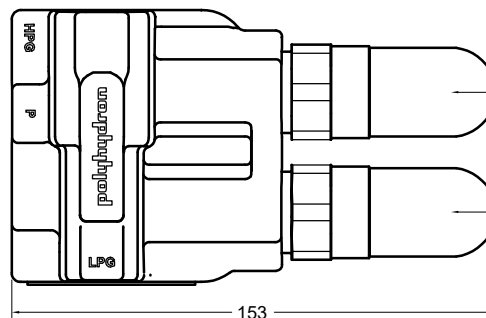
Dimensions in mm.



Hydraulic Symbol



Port details		
HP	:	G $\frac{3}{8}$
LP	:	G $\frac{3}{8}$
P	:	G $\frac{1}{2}$
HPG	:	G $\frac{1}{4}$
LPG	:	G $\frac{1}{4}$
T1	:	G $\frac{3}{8}$
T2	:	G $\frac{3}{8}$



High Pressure pump relief valve
 Pressure adjusting screw.

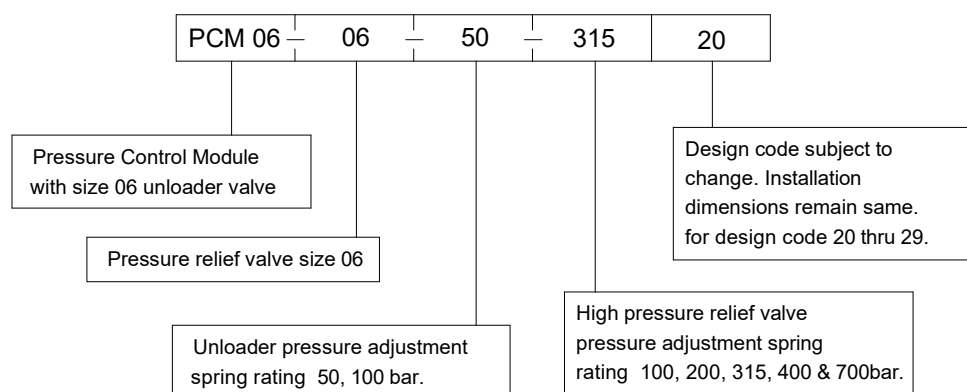
Low Pressure pump unloader valve
 Pressure adjusting screw.



Technical specifications

Construction	:	Direct acting poppet type high pr. relief valve Direct acting spool type low Pr. unloading valve
Mounting / Standard	:	Threaded body (Factory standard).
Mounting position	:	Optional.
Flow direction	:	As per Hydraulic Symbol.
Maximum operating pressure	:	For Low pressure pump 100 bar. For high pressure pump port 700 bar For port 'P' 700 bar.
Hydraulic medium	:	Mineral oil.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +70°C.
Fluid cleanliness requirement	:	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	:	High Pressure pump maximum flow : 25 lpm. Low Pressure pump maximum flow : 25 lpm
Mass (approx)	:	2.8 kg Approx.

Ordering code





A Polyhydron Group Company

PILOT OPERATED PRESSURE REDUCING VALVE Model : PPM

Ref. No. D 06281
Release 01 / 2026

ENGINEERING - 1 of 4

Description

Two stage pilot operated construction.

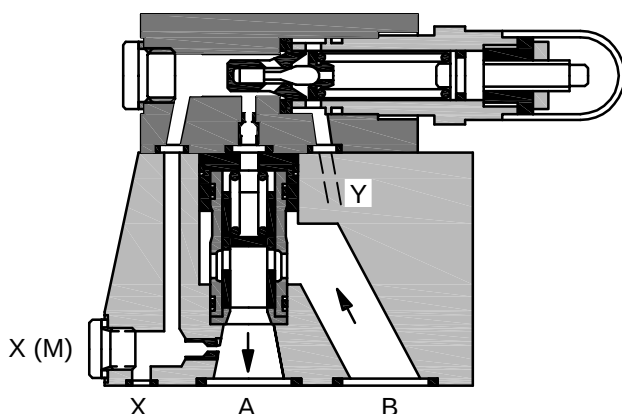
These valve are normally open type valve.

When the pressure in the secondary port (port A) increases to set pressure, the valve closes against the spring.

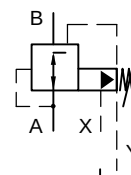
For reverse free flow, the valve can be supplied with a modular check valve (To be ordered separately).



Section



Hydraulic Symbol



Technical specifications

Construction	Two stage pilot operated valve.
Mounting	Subplate mounting - ISO 5781 Threaded bodies - Company standards
Mounting position	Optional.
Flow direction	From port B to port A.
Inlet pressure at port B	315 bar
Outlet pressure at port A.....	5 to 315 bar
Max. back pressure at port Y	60 bar
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +80°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Maximum flow handling capacity	For valve size 1080 l/min 20200 l/min 30300 l/min
Mass	Refer individual table.

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Unit Dimensions

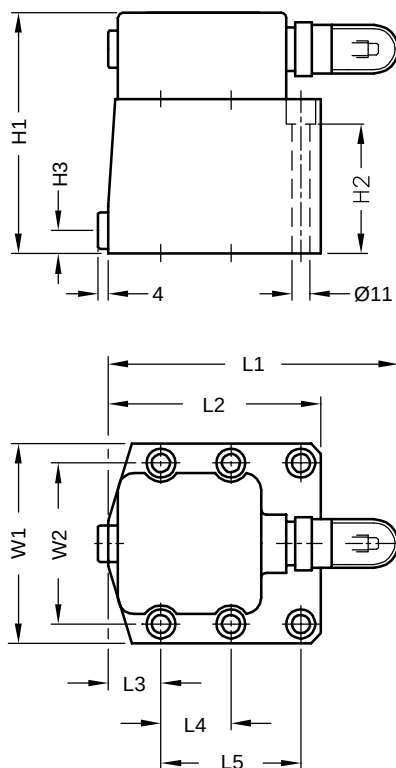
Dimensions in mm.

Valve : Subplate Type

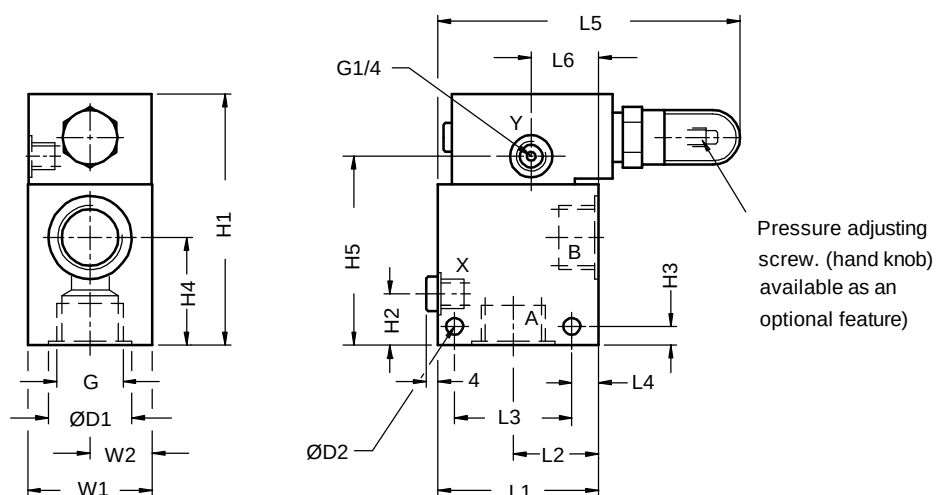


Size	10	20	30
Mass kg	5.5	7.7	13.5
H1	121	124	146
H2	75	75	75
H3	12	12	14
L1	155	157	170
L2	84	100	128
L2	31	28.9	31.8
L3	---	---	42.1
L4	42.9	60.3	84.1
W1	88	100	120
W2	66.7	79.4	96.8

Size	Valve fixing screws	Tightning torque
10	M10 x 90 L, 4 Nos.	70 Nm
20	M10 x 90 L, 4 Nos.	70 Nm
30	M10 x 90 L, 6 Nos.	70 Nm

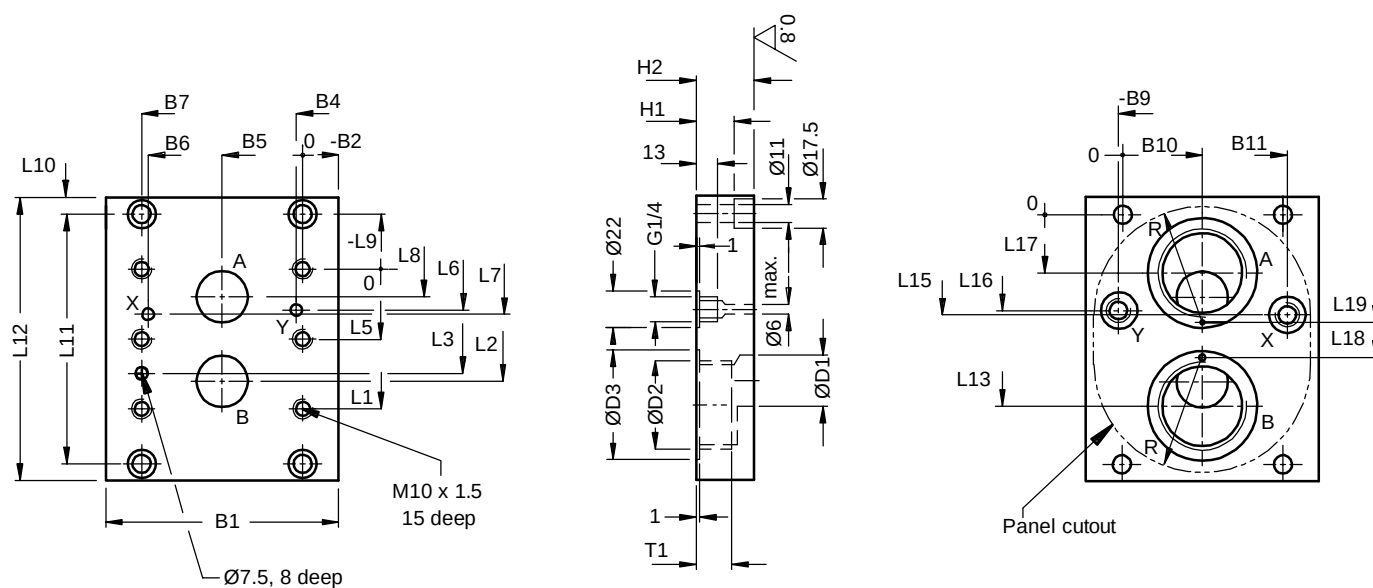


Valve : Threaded Type



Size	Mass (kg)	G	H1	H2	H3	H4	H5	L1	L2	L3	L4	L5	L6	W1	W2	ØD1	ØD2
20	9	G1	134	27	10	57	100	85	45	62	14	157	35.5	66	33	44	9
30	11	G1½	173	42	13	80	138	100	54	72	18	168	41.5	86	43	60	1

Subplate (Not in supply) details are only for reference.



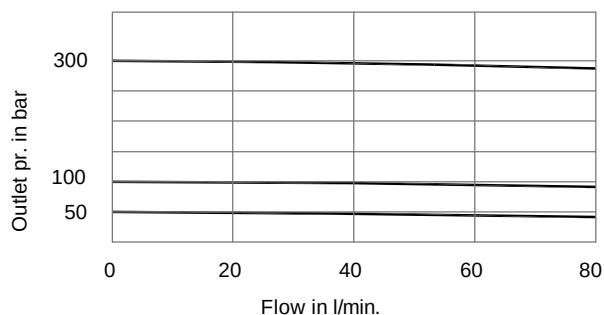
Size	Model	Mass kg	B1	B2	B4	B5	B6	B7	B9	B10	B11	ØD1	ØD2	ØD3	H2	H1	L1	L2
10	1SP06G03	2.4	86	9.7	7.9	33.3	58.7	66.7	-7.9	33.3	58.7	13	G3/8 G1/2	25 30	29	15	42.9	35.7
	1SR06G04																	
20	1SP08G06	5.2	112	16.3	6.4	39.7	73	79.4	3	39.7	75	22	G3/4 G1	36 44	45	25	60.5	49.2
	1SP08G08																	
30	1SP09G10	8.5	140	21.6	4	48.4	92.9	96.8	1.6	48.4	98.5	31	G1¼ G1½	54 60	47	35	84.1	67.5
	1SP09G12																	

Size	Model	L3	L5	L6	L7	L8	L9	L10	L11	L12	L13	L15	L16	L17	L18	L19	R	T1
10	1SP06G03	31.8	---	21.4	21.4	7.1	45.9	10	121	141	87	67.3	67.3	47	82	53	40	12
	89										45			14				
20	1SP08G06	44.5	---	39.7	20.8	11.1	22	16	104	136	77	42.6	64	27	57	47	52	16
	82										22			18				
30	1SP09G10	62.7	42.1	59.6	24.6	16.7	33	10	150	170	111	60	99.1	39	90	60	65	20
	115										35			22				

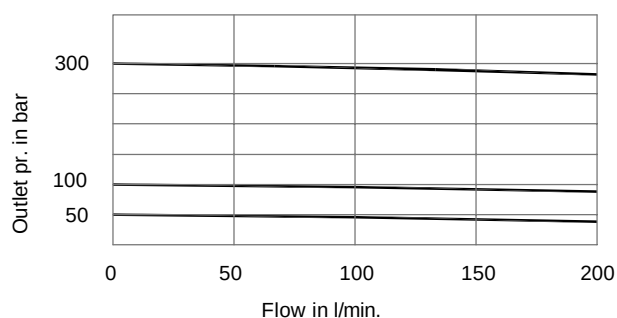


Performance Curves for PPM
(pressure related to flow, with 46 cSt at 40°C)

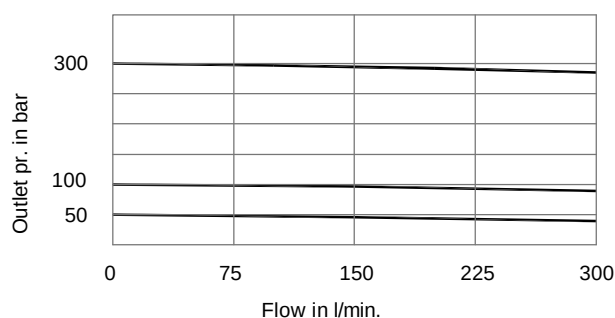
Size 10



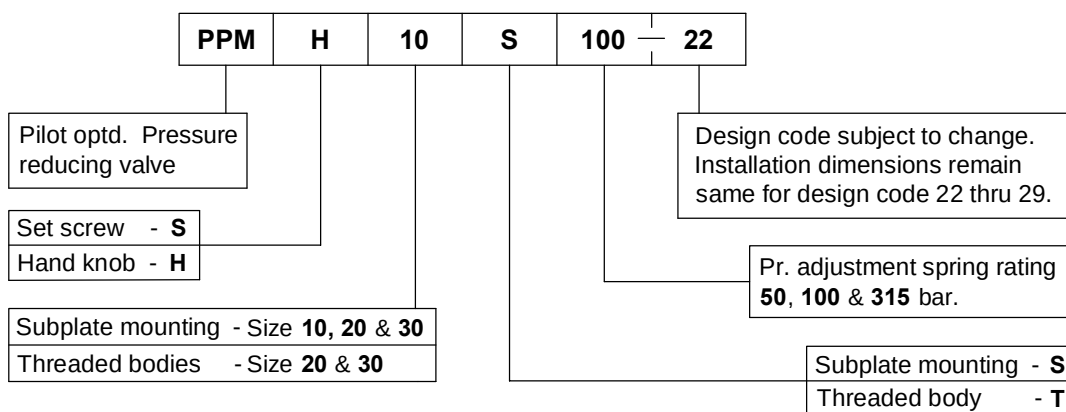
Size 20



Size 30



Ordering code





A Polyhydron Group Company

PILOT OPERATED PRESSURE RELIEF CUM UNLOADING VALVE Model : PPRU

Ref. No. D 03600

Release 01 / 2026

ENGINEERING - 1 of 4

Description

Two stage poppet type Relief cum unloading valve.

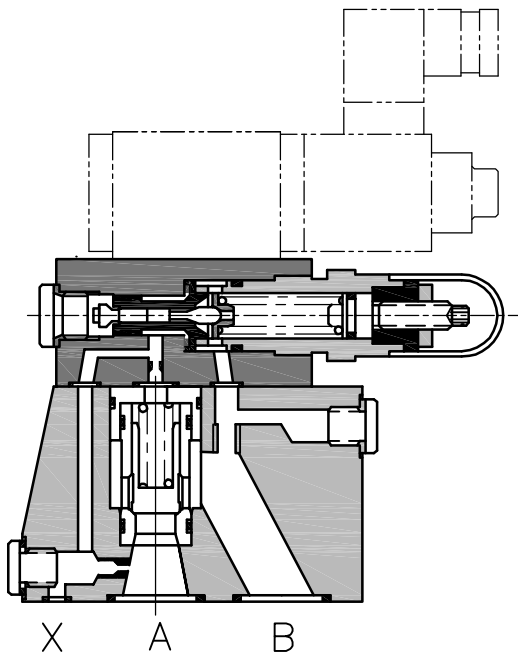
Additional solenoid unloading facility as an option feature.

Three standard sizes in Sub-plate mounting construction.

Two sizes in threaded mounting.



Section



Hydraulic Symbol

Description	Normally Open	Normally Close
Internal Pilot Internal Drain P - Unloading pressure		
X - External pilot Internal Drain P - Unloading pressure		
Internal Pilot Y - External Drain P - Unloading pressure		
X - External pilot Y - External Drain P - Unloading pressure		

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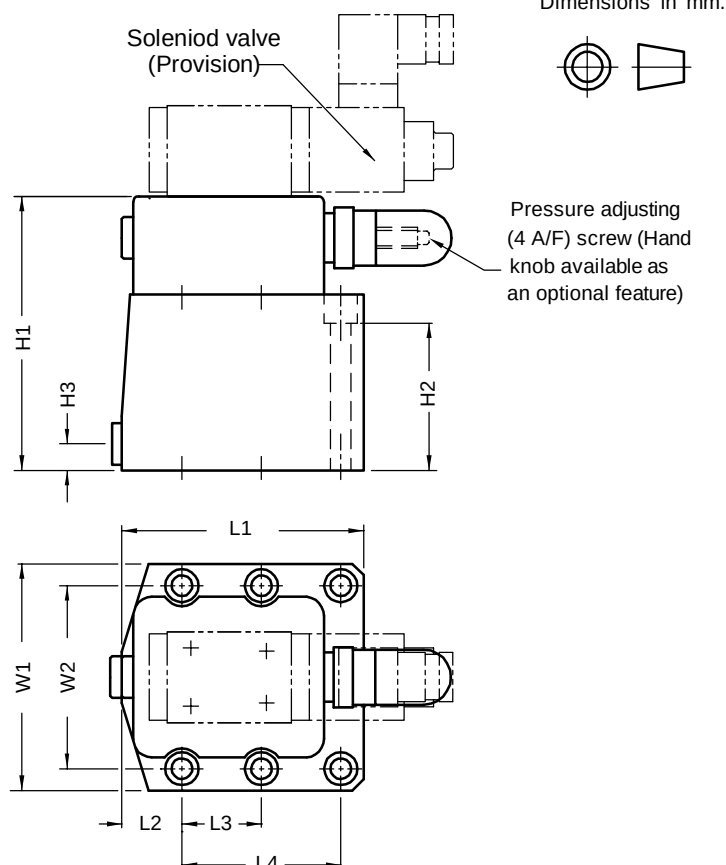
Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



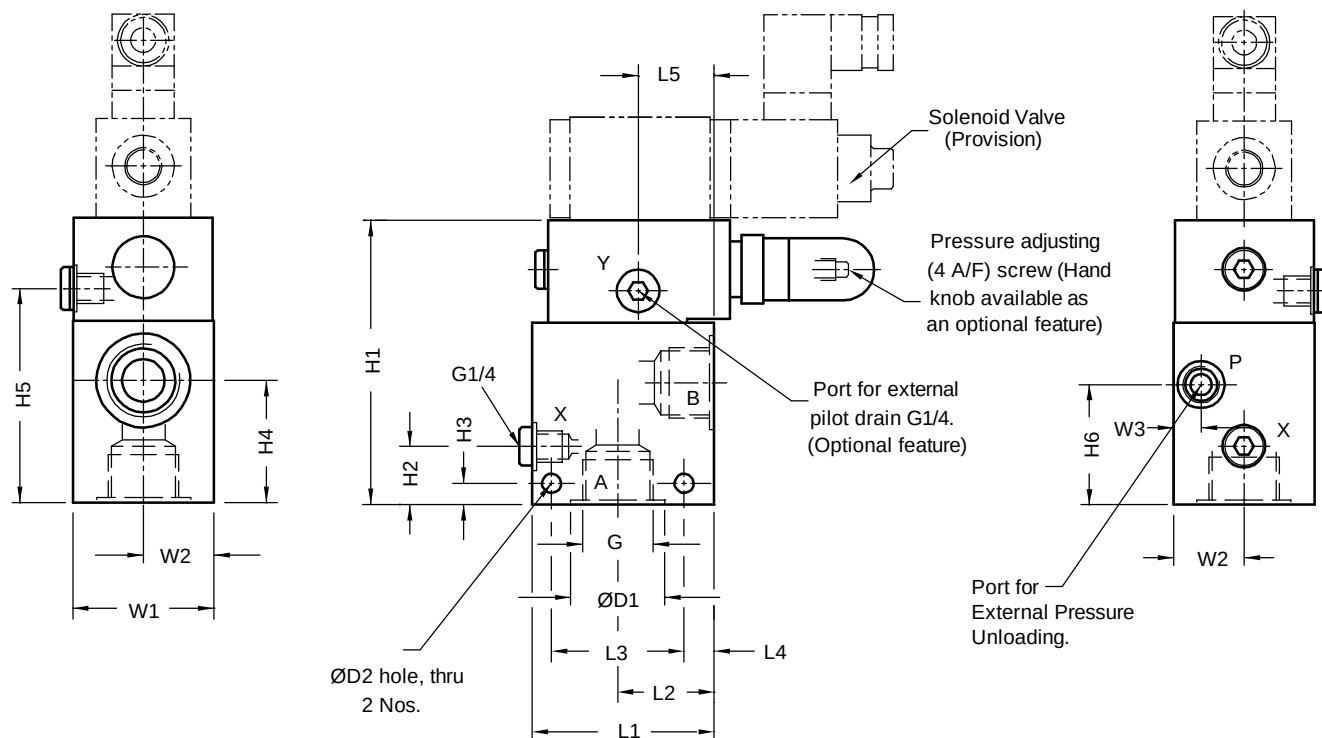
Unit Dimensions

Valve : Subplate Type

Size	10	20	30
Mass kg	4.8	6.7	12
H1	125	126	145
H2	95	75	75
H3	13	12	14
L1	84	100	128
L2	31	28.9	31.8
L3	---	---	42.1
L4	42.9	60.3	84.1
W1	88	100	120
W2	66.7	79.4	96.8



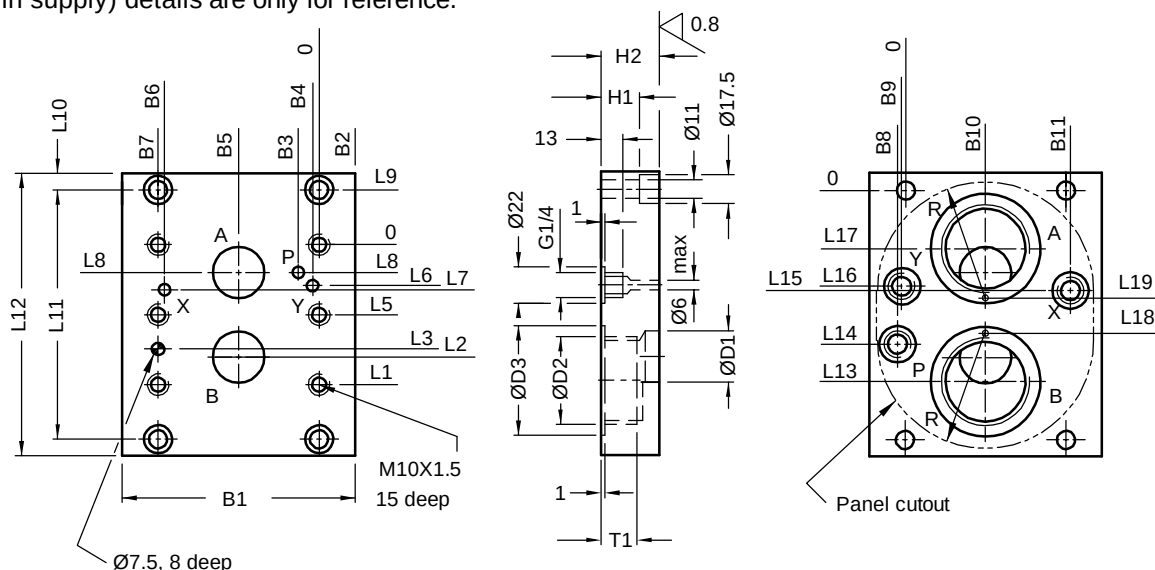
Valve : Threaded Type



Size	Mass (kg)	G	H1	H2	H3	H4	H5	H6	L1	L2	L3	L4	L5	W1	W2	W3	ØD1	ØD2
20	4.8	G1	138	27	10	57	100	50	85	45	62	14	35.5	66	33	12	44	9
30	9.5	G1½	175	42	13	80	138	80	100	54	72	18	41.5	86	43	14	60	11



Subplate (Not in supply) details are only for reference.



Size	Model No.	Mass kg	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	ØD1	ØD2	ØD3
10	1SU06G02	2.4	86	9.6	11	7.9	33.3	58.7	66.7	-33.3	-7.9	33.3	58.7	13	G1/4	22
	1SU06G03														G3/8	25
	1SU06G04														G1/2	30
20	1SU06G06	5.2	112	16.3	14	6.4	39.7	73	79.4	3	3	39.7	75	22	G3/4	36
	1SU06G08														G1	44
30	1SU06G10	8.5	140	21.6	15	4	48.4	92.9	96.8	1.6	1.6	48.4	98.5	31	G1¼	54
	1SU06G12														G1½	60

Size	Model No.	H1	H2	L1	L2	L3	---	L5	L6	L7	L8	L9	L10	L11	L12	L13
10	1SU06G02	15	29	49.2	35.7	31.8	---	---	21.4	21.4	7.1	45.9	10	121	141	85
	1SU06G03															
	1SU06G04															
20	1SU06G06	25	45	60.5	49.2	44.5	---	---	39.7	20.8	11.1	22	16	104	136	77
	1SU06G08															
30	1SU06G10	35	47	84.1	67.5	62.7	---	42.1	59.6	24.6	16.7	33	10	150	170	111
	1SU06G12															

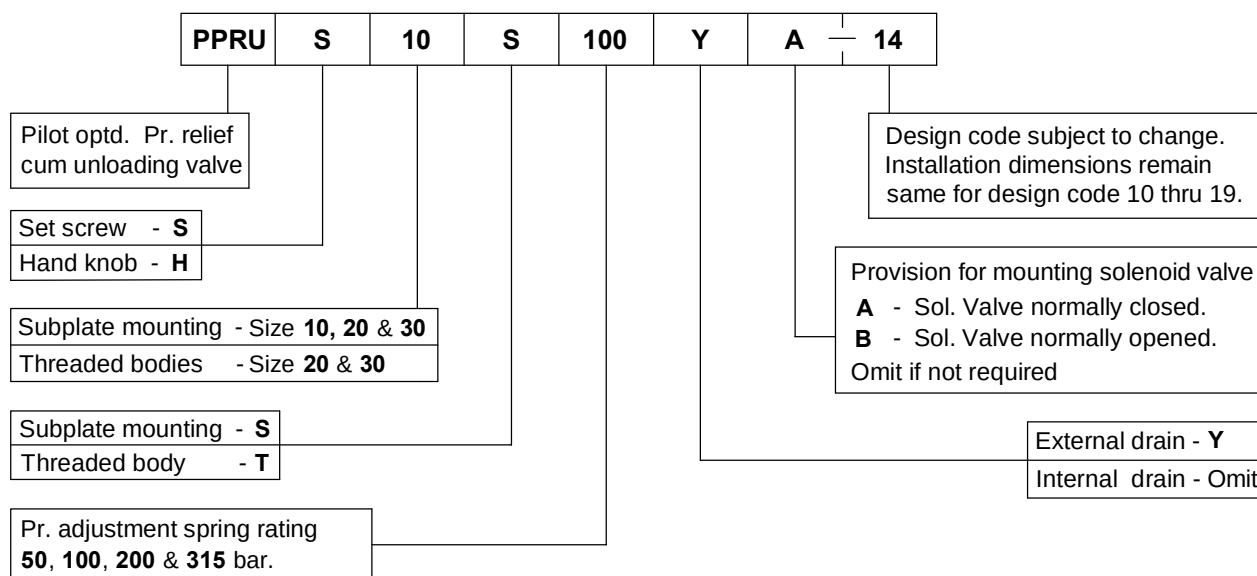
Size	Model No.	L14	L15	L16	L17	L18	L19	R	T1	Valve fixing screw	Tightning torque
10	1SU06G02	11	67.3	67.3	49	82	53	40	12	SHCS M10 x 110L 4 nos	70 Nm
	1SU06G03									SHCS M10 x 90L 4 nos	
	1SU06G04									SHCS M10 x 90L 6 nos	
20	1SU06G06	39	42.6	64	27	57	47	52	16	SHCS M10 x 90L 4 nos	70 Nm
	1SU06G08									SHCS M10 x 90L 6 nos	
30	1SU06G10	50	60	99.1	39	90	60	65	20	SHCS M10 x 90L 6 nos	70 Nm
	1SU06G12									SHCS M10 x 90L 6 nos	



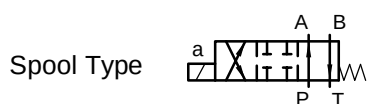
Technical specifications

Construction	Two stage poppet valve.
Mounting	Subplate mounting - ISO 5781
	Threaded bodies - Company standards
	with additional pilot pressure port 'P'.
Mounting position	Optional.
Flow direction	From port A to port B.
Maximum Operating pressure	
For port A ,P and X	315 bar.
For port B and Y	60 bar..
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +80°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Maximum flow handling capacity	For valve size
	10100 l/min
	20200 l/min
	30400 l/min
Mass	Refer individual table.

Ordering code



Note : Solenoid (refer spool type) to be ordered separately.





A Polyhydon Group Company

PILOT OPERATED PRESSURE RELIEF VALVE Model : PPR

Ref. No. D 04897
Release 01 / 2025

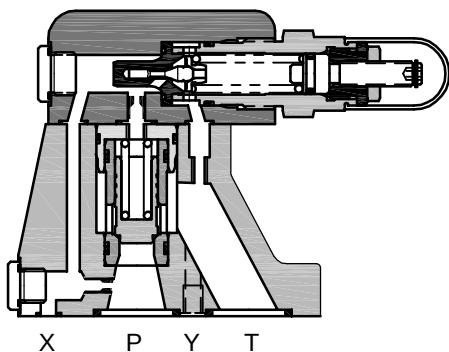
ENGINEERING - 1 of 3

Description

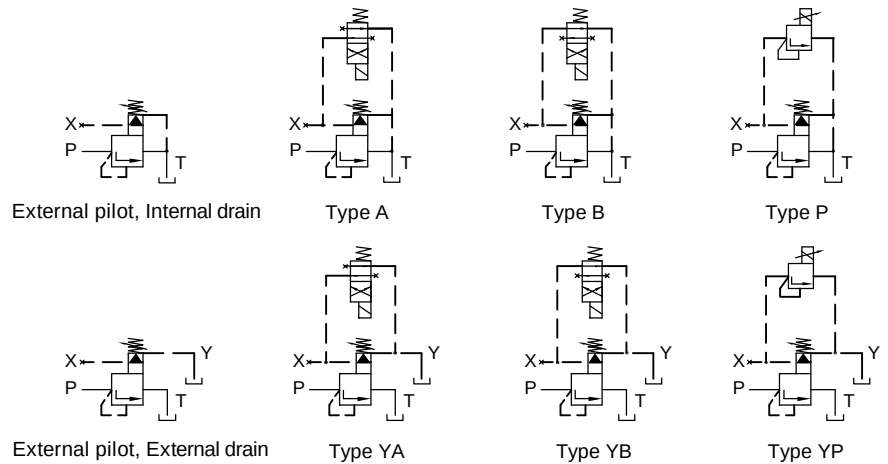
Two stage poppet type Relief valve.
Solenoid unloading or proportional relieving facility as an optional feature.
Vent connection for remote control.
Two standard sizes in Threaded construction.
Three standard sizes in Sub-plate mounting construction.



Section



Hydraulic symbol



Unit dimensions

Valve : Threaded body

Dimensions in mm.

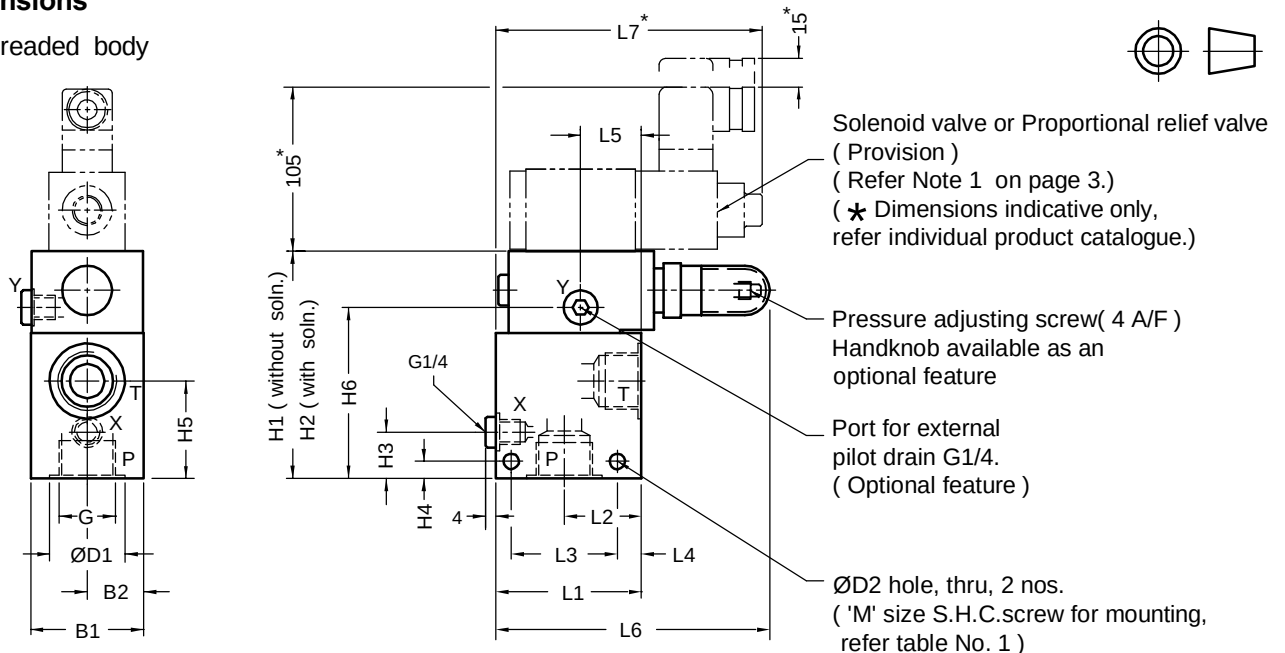


Table No. 1

Size	Mass kg.	G	H1	H2	H3	H4	H5	H6	L1	L2	L3	L4	L5	L6	L7	B1	B2	ØD1	ØD2	M
20	4.8	G1	131	140	25	10	57	100	85	43	62	12	33.5	153	155	66	33	44	9	M8
30	9.5	G1-1/2	170	178	42	13	80	138	100	54	72	18	41.5	162	163	86	43	60	11	M10

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Valve : Subplate mounting

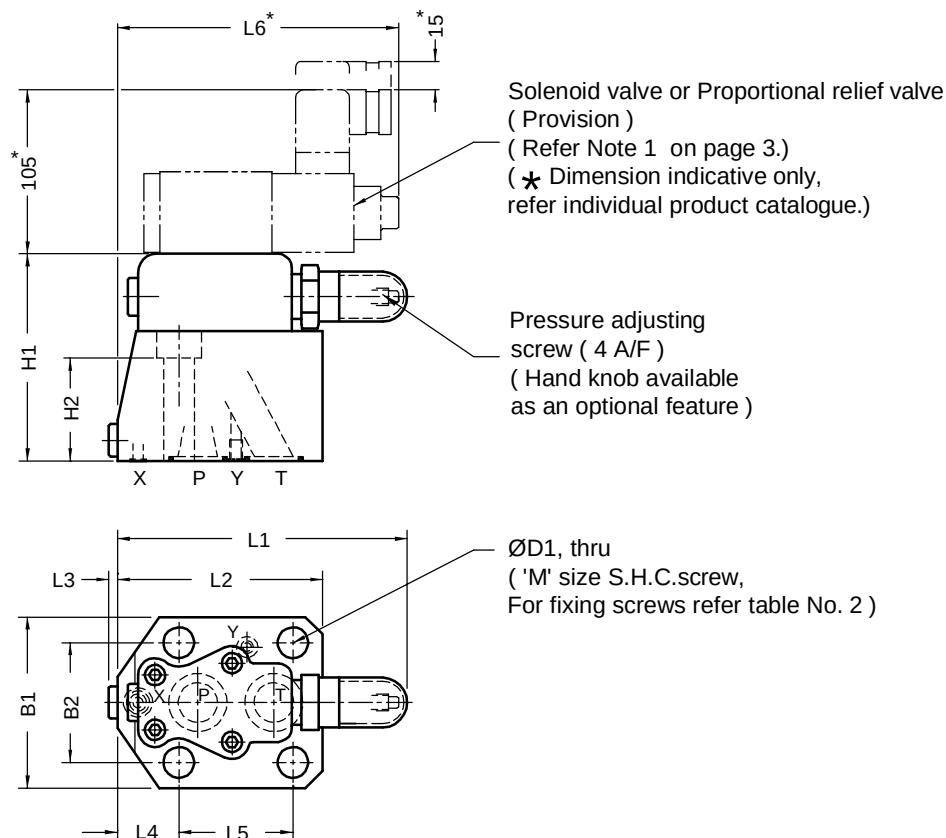
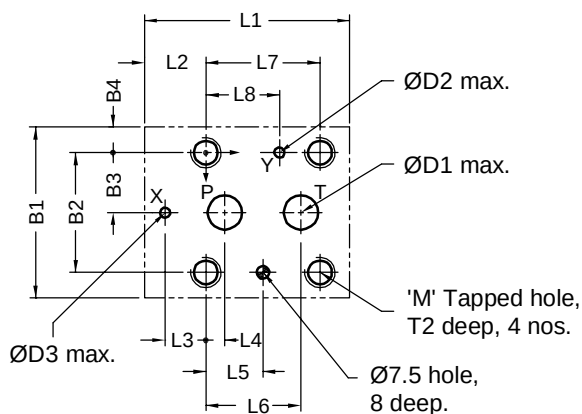


Table No. 2

Size	H1	H2	L1	L2	L3	L4	L5	L6	ØD1	B1	B2	Mass kg	M Valve fixing Screws	Tightening Torque
10	120	82	150	80	4	12.9	53.8	142	13	80	53.8	4.8	M12 x 100 L - 4 nos.	125 Nm
20	115	66	167	120	4	35.9	66.7	163	18	100	70	6.7	M16 x 90 L - 2 nos. M16 x 50 L - 2 nos.	300 Nm
30	143	93	182	155	4	47.3	88.9	178.5	20	120	82.6	12	M18 x 120 L - 4 nos.	500 Nm



Interface details

Note : Surface roughness : Ra value is equal to or better than 0.8 µm
Surface flatness : 0.01 - 100 mm.

Table No. 3

Size	L1	L2	L3	L4	L5	L6	L7	L8	B1	B2	B3	B4	ØD1	ØD2	ØD3	M	T2
10	80	12.9	0	22.1	22.1	47.5	53.8	31.8	80	53.8	26.9	13.1	13	4.8	4.8	M12	22
20	120	35.9	23.8	11.1	33.4	55.6	66.7	43.7	100	70	35	15	23	6.3	6.3	M16	26
30	155	47.3	31.8	12.7	44.5	76.2	88.9	54.9	120	82.6	41.3	18.7	32	6.3	6.3	M18	30

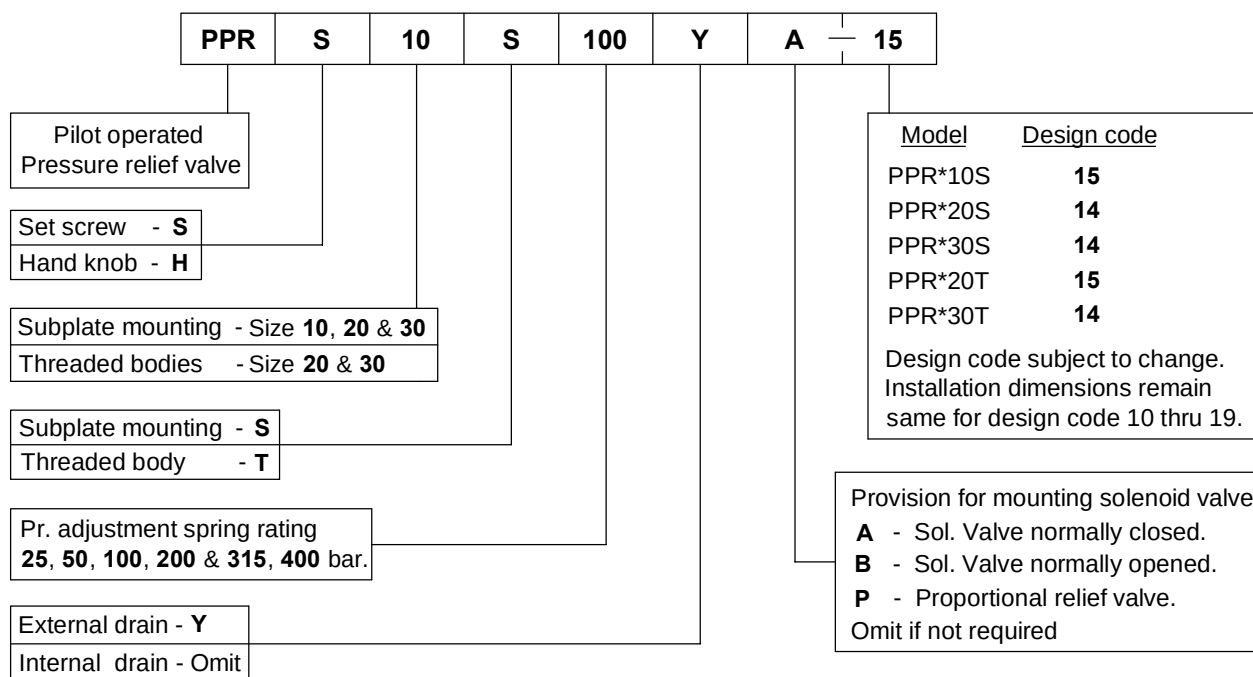


Technical specifications

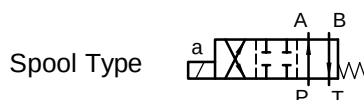
Construction	Two stage poppet valve.
Mounting	Threaded bodies - Company standards
	Subplate mounting - ISO6264.
Mounting position	Optional.
Flow direction	From port P to port T.
Maximum Operating pressure	
For port P and X	315 bar.
For port T and Y	210 bar.
(For model with solenoid valve mounting provision)	
For port T and Y.....	Max. pressure is equal to tank port pressure rating of the valve mounted. ★
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +70°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Maximum flow handling capacity	For valve size
	10100 l/min
	20200 l/min
	30400 l/min
Mass	Refer individual table.

★ Port pressure is additive to the setting of the valve.

Ordering code



Note : Solenoid (refer spool type) and Subplate to be ordered separately.





A Polyhydron Group Company

PILOT OPTD. PRESSURE RELIEF VALVE MODULAR CONSTRUCTION

MPPR *0601**

Ref. No.D 09252
Release : 12 / 2025

ENGINEERING - 1 of 2

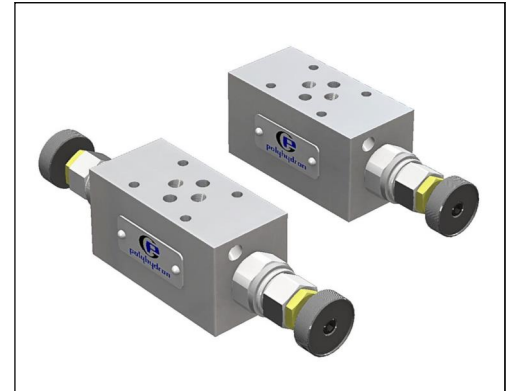
Description

Pilot optd. pressure relief valve modular construction for vertical stacking assemblies.

For Cartridge Valve details, refer data sheet - D 23100 A

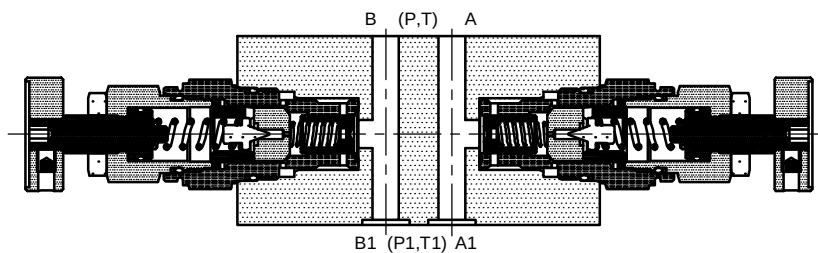
Interface conforming to
ISO 4401 - AB - 03 - 04 - A
IS 10187
DIN 24340

Available in four models.

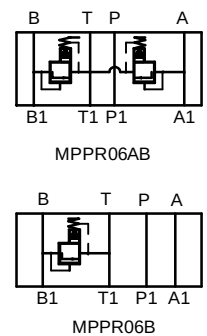
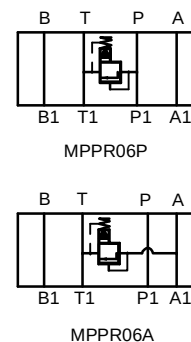


Section

Model : **MPPR * 06 AB * 01**



Hydraulic symbol



Technical specifications

Construction	Pilot operated spool type.
Mounting / Standard	Threaded Cartridge (Factory standard)
Mounting position	Optional.
Flow direction	From P to T.
Maximum operating pressure.....	For port P 350 bar For port T 60 bar For port Y 60 bar
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +80°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	30 lpm (Refer data sheet no. D 23100 A).
Mass (approx)	1.5 kg.

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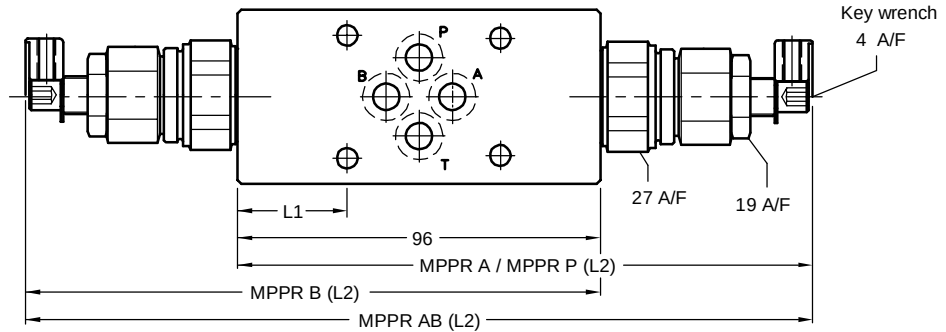
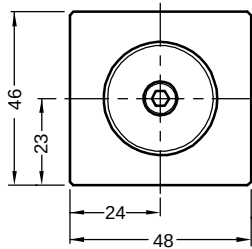
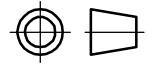
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Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



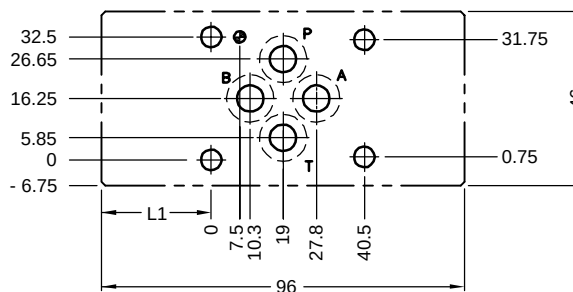
Unit dimensions

Dimensions in mm.



INTERFACE

Model	L1	L2
MPPR AB	29	208
MPPR A	16	152
MPPR B	39.5	152
MPPR P	18	152



Note

Interface Surface roughness : **Ra** value should be equal to or better than 0.8 μm

Interface Surface flatness : 0.01 - 100 mm.

Ordering code

MPPR	S	06	AB	100	01
-------------	----------	-----------	-----------	------------	-----------

Modular construction
Pilot optd. pr. relief valve.

Adjustable pr. with setscrew

S

Adjustable pr. with handknob

HSize **06**Design code subject to change.
Installation dimensions remain
same for design code thru
01 thru **09**Pr. Rating - **50,100,200 & 315**

For Modular construction only

AB	Cartridge on A & B ports.
A	Cartridge on A port.
B	Cartridge on B port.
P	Cartridge on P port.



A Polyhydron Group Company

PILOT OPERATED PRESSURE REDUCING VALVE PPM *06K*10

Ref. No.D 23100 B
Release : 06 / 2026
ENGINEERING - 1 of 2

Description

Pilot operated spool type Pressure Reducing valve cartridge.

Low pressure over-ride, due to larger flow area.

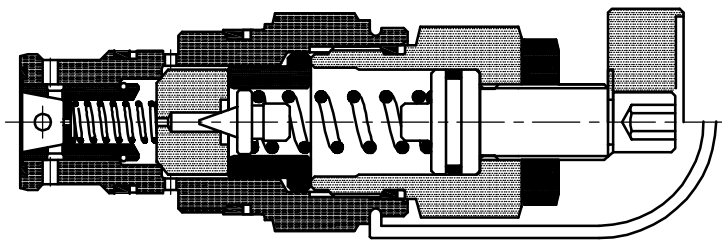
Two options for pressure adjustments,

- With set screw and lock nut.
- With hand knob and lock nut.

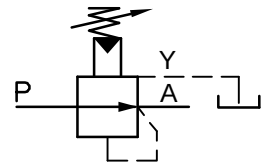


Section

Valve : PPM

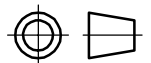
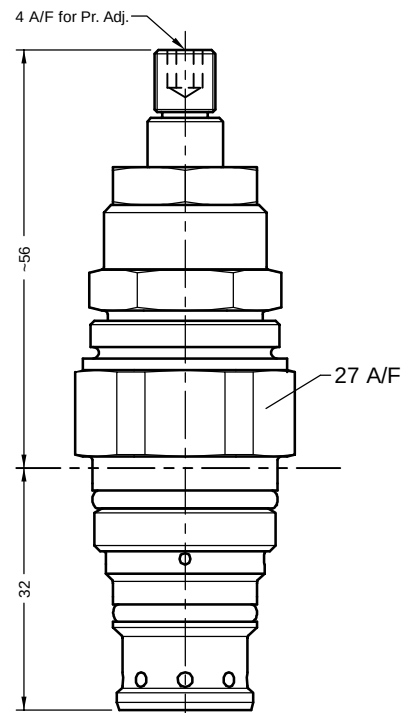
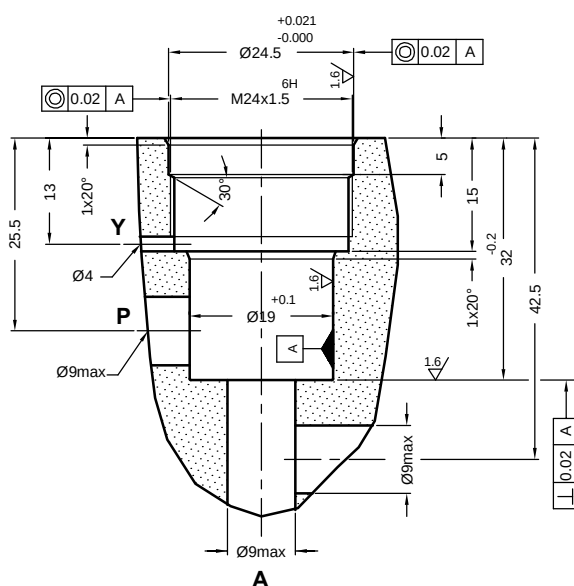


Hydraulic symbol



Unit dimensions

Dimensions in mm.



polyhydron pvt. ltd.

78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

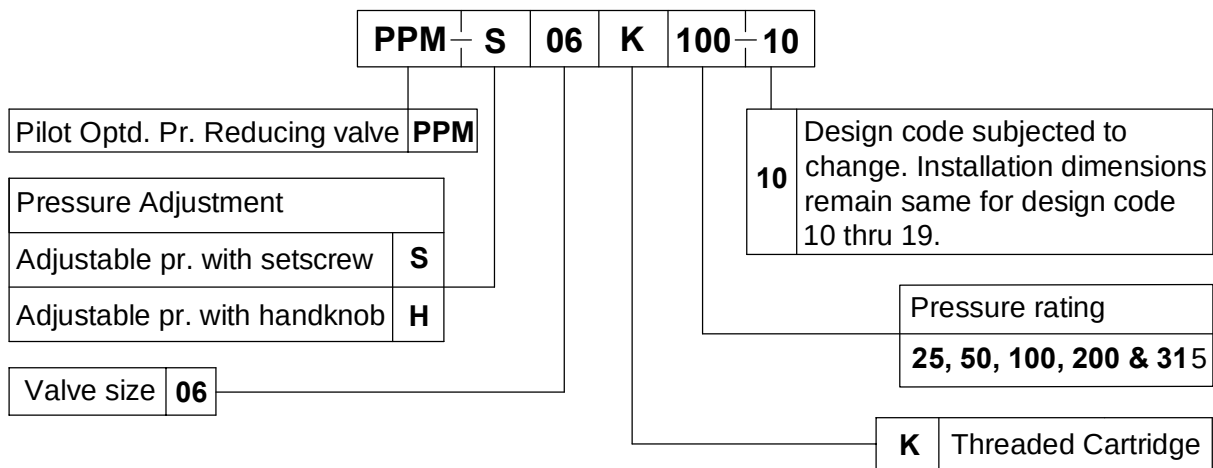
Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



Technical specifications

Construction	Pilot operated spool type.
Mounting / Standard	Threaded Cartridge (Factory standard)
Mounting position	Optional.
Flow direction	From P to A.
Maximum operating pressure.....	For port P, A, B 315 bar For port Y 60 bar
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +80°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	30 lpm.
Mass (approx)	0.2 kg.

Ordering code





Ref. No.D 23100 A
Release : 06 / 2026

Description

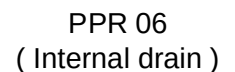
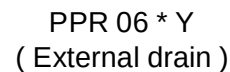
Pilot operated spool type relief valve cartridge.

Low pressure over-ride, due to larger flow area.

Two options for pressure adjustments,

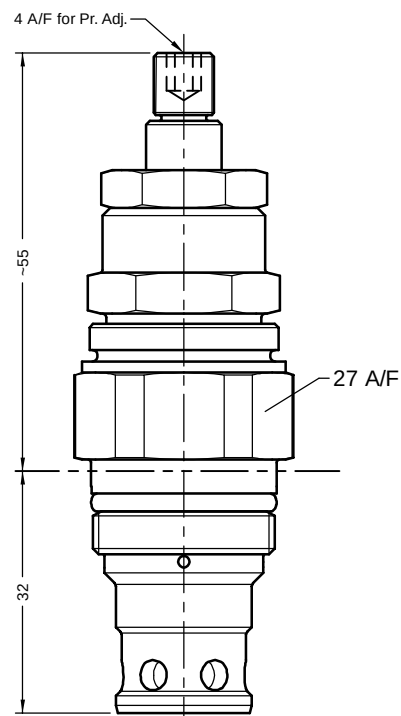
- With set screw and lock nut.
- With hand knob and lock nut.

Internal drain valve can be converted into external drain valve.

Valve : **PPR**

Unit dimensions

Dimensions in mm.



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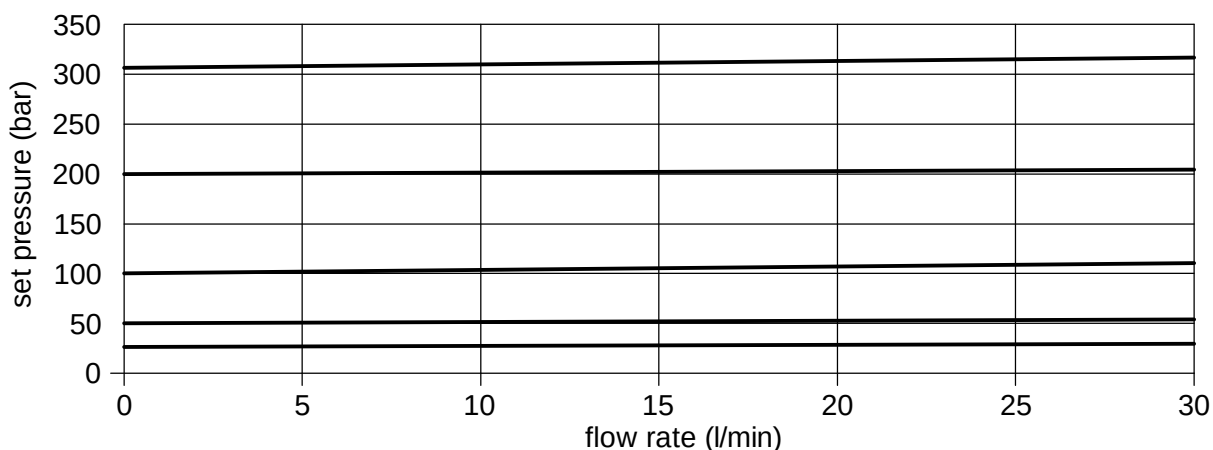
Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



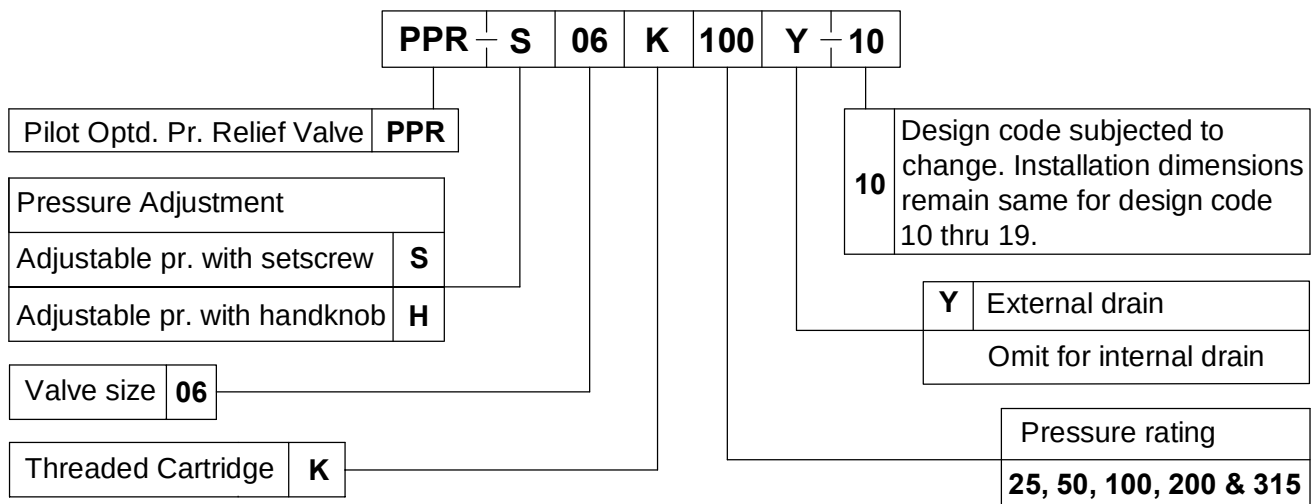
Technical specifications

Construction	Pilot operated spool type.
Mounting / Standard	Threaded Cartridge (Factory standard)
Mounting position	Optional.
Flow direction	From P to T.
Maximum operating pressure.....	For port P 350 bar For port T 60 bar For port Y 60 bar
Hydraulic medium	Mineral oil.
Viscosity range	10 cSt to 380 cSt
Fluid temperature range	-20°C to +80°C
Fluid cleanliness requirement	ISO 4406 20/18/15 or better.
Nominal flow handling capacity	refer performance curve.
Mass (approx)	0.2 kg.

Performance Curves for PPR06K
(pressure related to flow, with 46 cSt at 40°C)



Ordering code





A Polyhydron Group Company

DIRECT ACTING PRESSURE SEQUENCE VALVE Model : DPS *06S ***

Ref. No. D 04505
Release: 11 / 2025

ENGINEERING - 1 of 3

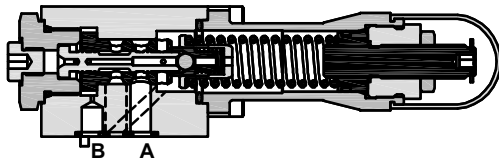
Description

Direct acting, spooltype pressure sequence valve.
Four pressure rating 25, 50, 100 and 200 bar with primary pressure rating to 315 bar.

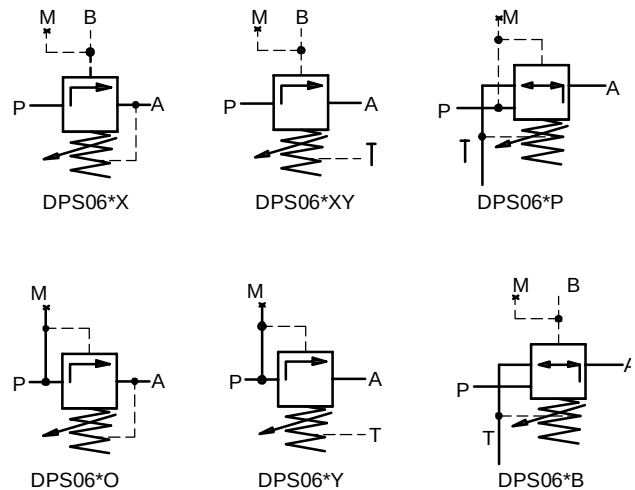
Pressure setting by screw of hand knob adjustment.



Section



Hydraulic symbol



Technical specifications

Construction	:	Direct acting spool type.
Mounting style	:	Subplate mounting.
Mounting interface	:	Sub-plate mounting - As per ISO 6403.
Flow direction	:	From Port P to Port A.
Working pressure	:	For Port P and Port B.....315 bar. For Port A.....60 bar (max.) for model DPS*06SO** and DPS*06SX** only, 315 bar for remaining models. For Port T.....60 bar (max). (Pressure at Port A of models DPS*06SO** and DPS*06SX** will increase the set value of the valve by the same amount, while handling the flow.)
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Max. flow handling capacity	:	Refer graph
Mass approx.	:	1.54 Kg.

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78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
E-mail : sales@polyhydron.com
Website : www.polyhydron.com



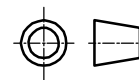
Technical drawing of the 1000 Series pressure washer, showing side and front views with dimensions and specifications.

Side View Dimensions:

- Overall length: 178
- Height at handle: 47
- Height at trigger gun: 25
- Trigger gun: 22 A/F
- Nozzle: 5 A/F
- Specification: Ø5.5 thru, 4 Nos. (Tightening torque 6 Nm)

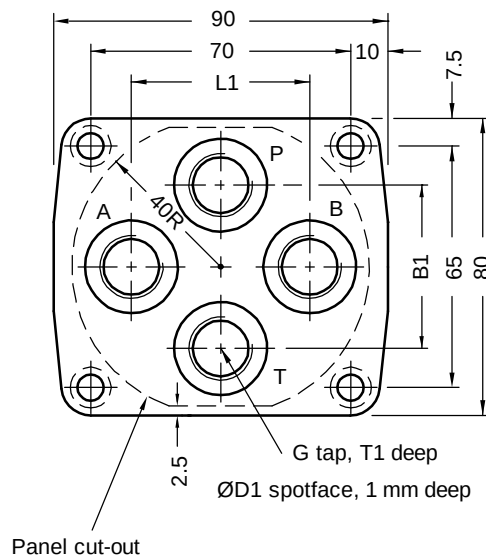
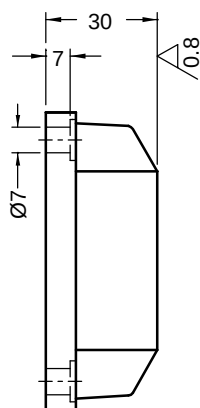
Front View Dimensions:

- Overall width: 48 (Ø48)
- Overall length: 180
- Distance from handle to trigger gun: 12
- Distance from trigger gun to nozzle: 40.5
- Distance from handle to nozzle: 60
- Distance from handle to trigger gun: 8
- Distance from trigger gun to nozzle: 31
- Distance from handle to trigger gun: 32.5
- Distance from handle to trigger gun: 44
- Pressure gauge port



Technical drawing of a mechanical part (Fig. 10.10) showing a top view. The part is a rectangular plate with rounded corners and a central rectangular cutout. The drawing includes the following dimensions and features:

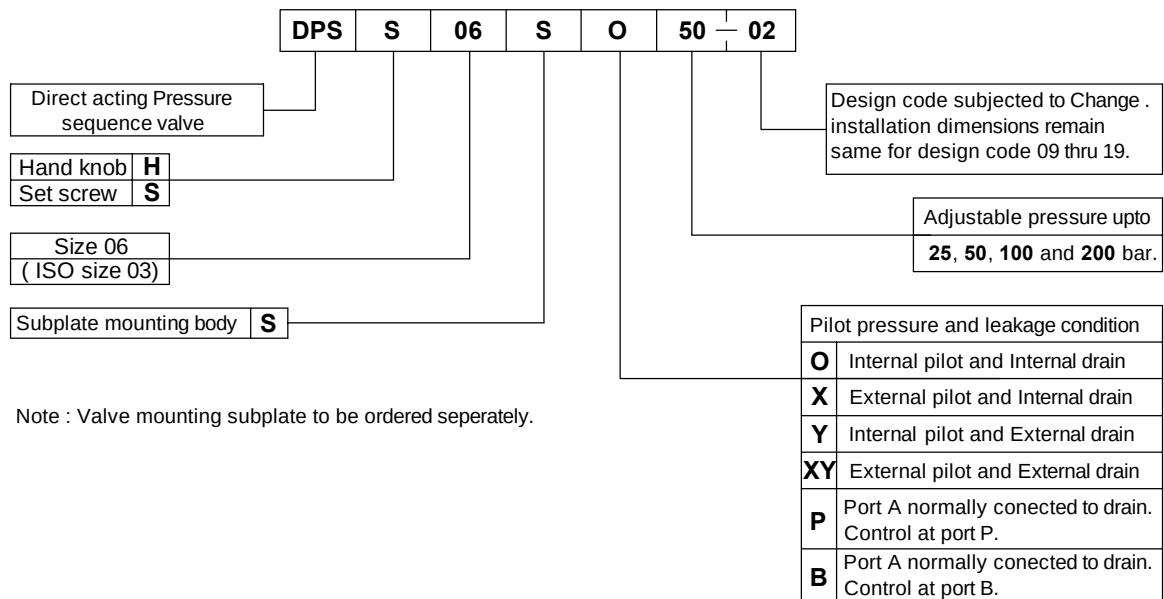
- Overall Dimensions:**
 - Width: 32.5
 - Height: 31.75
 - Inner Cutout Width: 26.65
 - Inner Cutout Height: 16.25
- Corner Features:**
 - Four corner holes: Ø4 drill, 5 deep.
 - Four corner slots: M5 tap, 9 deep, 4 Nos.
- Central Features:**
 - Central hole: Ø6.3 drill, 4 Nos.
 - Four small holes labeled A, B, P, and T.
 - Four small holes labeled 0, 7.5, 10.3, and 19.
- Other Dimensions:**
 - 16.25 (Distance from top edge to center of corner holes)
 - 0 (Distance from top edge to center of central hole)
 - 0.75 (Distance from top edge to center of corner slots)
 - 5.85 (Distance from top edge to center of corner slots)
 - 16.25 (Distance from bottom edge to center of corner holes)
 - 26.65 (Distance from bottom edge to center of corner slots)
 - 31.75 (Distance from bottom edge to center of corner slots)



Subplate	B1	ØD1	G	L1	T1
1SD03G02	34	22	G1/4	36	12
1SD03G03	44	25	G3/8	48	12

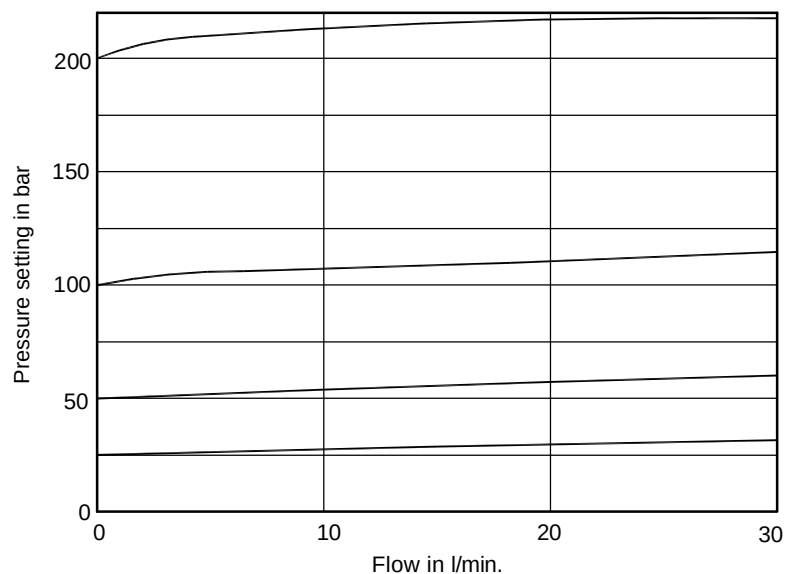


Ordering code



Performance graph

Oil used : ISO VG 68, Viscosity : 68 cSt at 40 °C, Test temperature : 52 - 55 °C.



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Due to continuous improvement in the design of the product, the actual product supplied may look different than shown above.

For critical applications, please ask for certified installation drawing.

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Subject to revision



A Polyhydron Group Company

DIRECT ACTING PRESSURE REDUCING VALVE Model : DPM*06S***

Ref. No. D 04479

Release: 11 / 2025

ENGINEERING - 1 of 3

Description

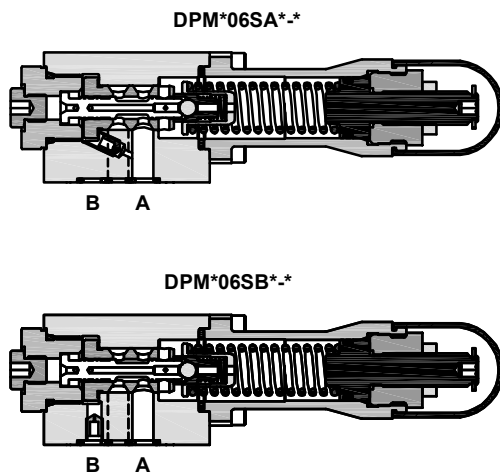
Direct acting, spooltype pressure reducing valve with control of pressure from either secondary circuit (Model DPM*06SA) or from a remote source (Model DPM*06SB).

Four pressure rating 25, 50, 100 and 200 bar with primary pressure rating to 315 bar.

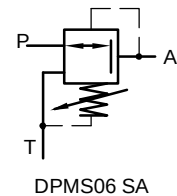
Pressure setting by screw of hand knob adjustment.



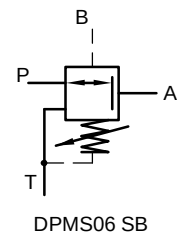
Section



Hydraulic symbol



DPMS06 SA



DPMS06 SB

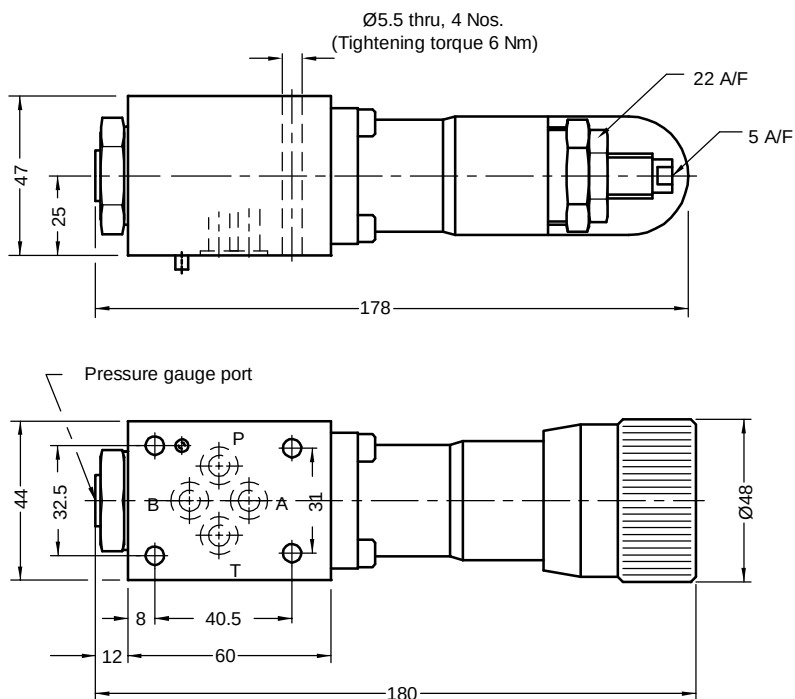
Technical specifications

Construction	:	Direct acting spool type.
Mounting style	:	Subplate mounting.
Mounting interface	:	Sub-plate mounting - As per ISO 6403.
Flow direction	:	Free flow from Port P to Port A.
Working pressure	:	For Port P.....315 bar. For Port A and Port B.....200 bar. For Port T.....60 bar (max).
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Max. flow handling capacity	:	Refer graph
Mass approx.	:	1.54 Kg.

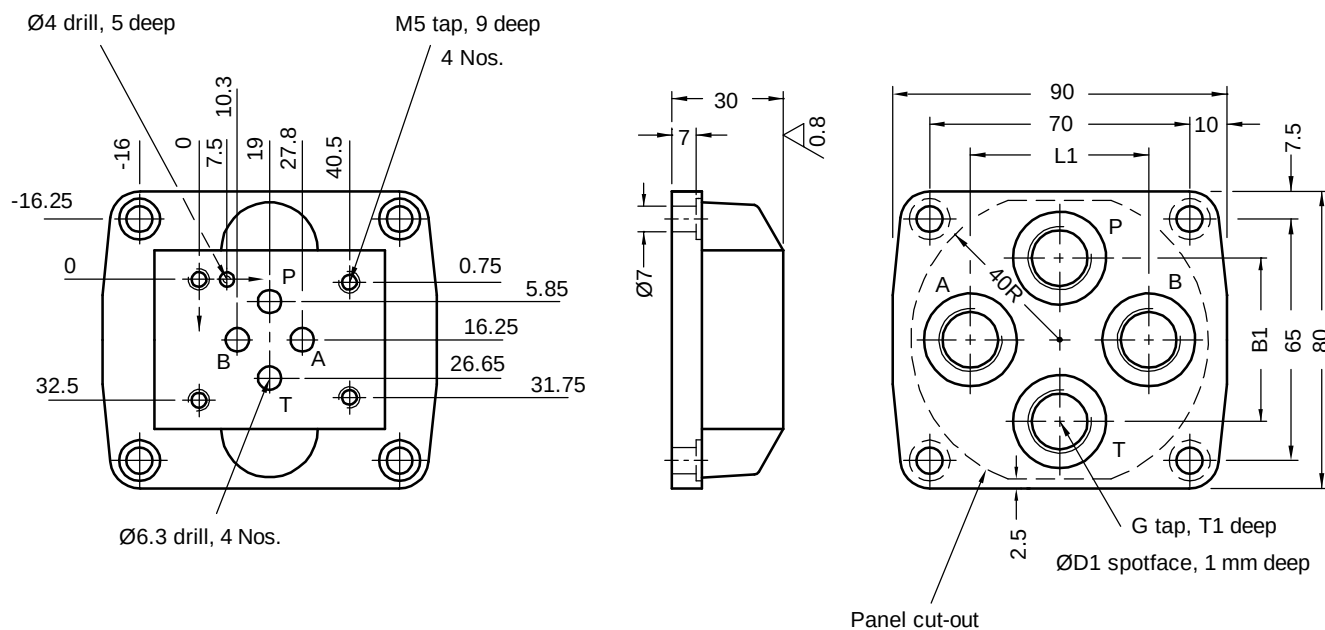


Unit dimensions

Dimensions in mm.



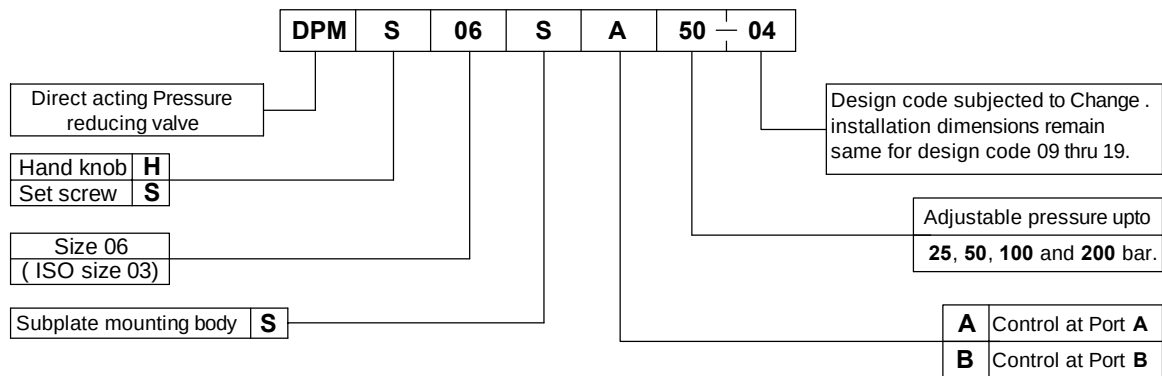
Valve mounting Sub plate



Subplate	B1	ØD1	G	L1	T1
1SD03G02	34	22	G1/4	36	12
1SD03G03	44	25	G3/8	48	12



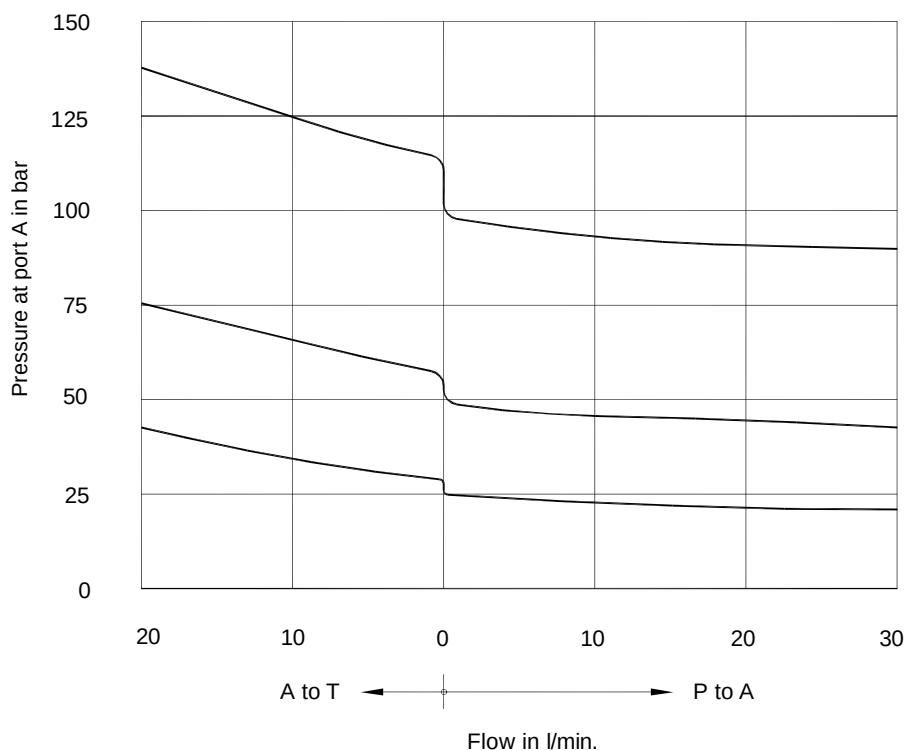
Ordering code



Note : Valve mounting subplate to be ordered seperately.

Performance graph

Oil used : ISO VG 68, Viscosity : 68 cSt at 40 °C, Test temperature : 52 - 55 °C.



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For critical applications, please ask for certified installation drawing.



A Polyhydon Group Company

DIRECT ACTING PRESSURE RELIEF VALVE Model : DPR

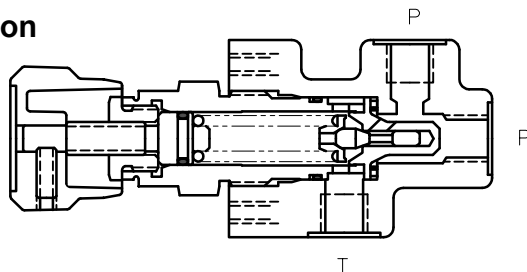
Ref. No. D 04896
Release: 01 / 2026
ENGINEERING - 1 of 5

Description

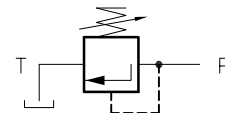
- Direct acting valve.
Guided poppet design.
Cushion arrangement for greater stability and noise control.
- Three sizes - 06, 10 and 20.
Three mounting style - Threaded cartridge.
Threaded port body.
Subplate Type (Factory std. and ISO std.)
- Six pressure ranges - Up to 25, 50, 100, 200, 315 & 400.
700 bar. (In size of 06 only.)
- Two options for pressure - With set screw and lock nut.
adjustment With hand Knob.



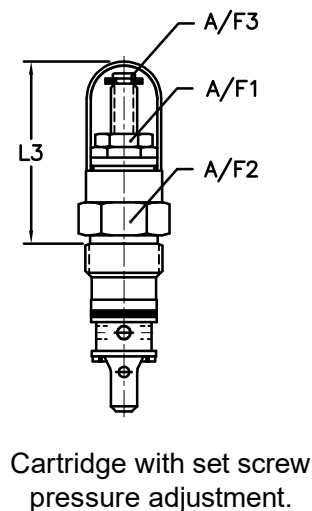
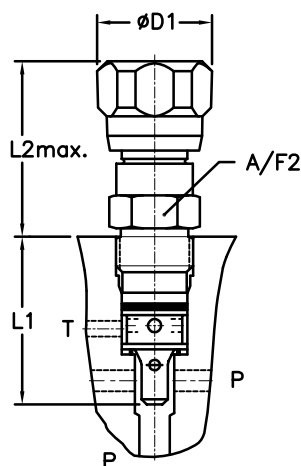
Section



Hydraulic Symbol

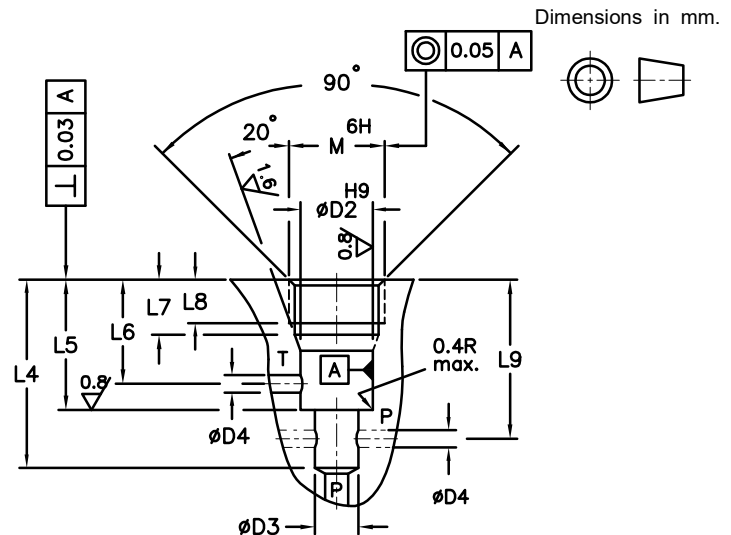


Unit dimensions



Cartridge with set screw
pressure adjustment.

Cartridge with hand knob
for pressure adjustment.



Cartridge cavity machining details.

Size	Mass kg	A/F1	A/F2	A/F3	6H M	ØD1	6H ØD2	ØD3	ØD4	L1	L2	L3	±0.5 L4	±0.2 L5	±0.2 L6	±0.2 L7	L8	L9	Tightening torque Nm
06	0.45	19	30	4	M28 x 1.5	43	25	15	6	64	64	65	65	45	35	19	15	55	120
10	0.9	19	36	5	M35 x 1.5	48	32	19	10	77	82	80	80	52	41	23	18	69	140
20	1.6	19	46	5	M45 x 1.5	60	40	24	20	103	85	79	110	70	54	27	21	92	170

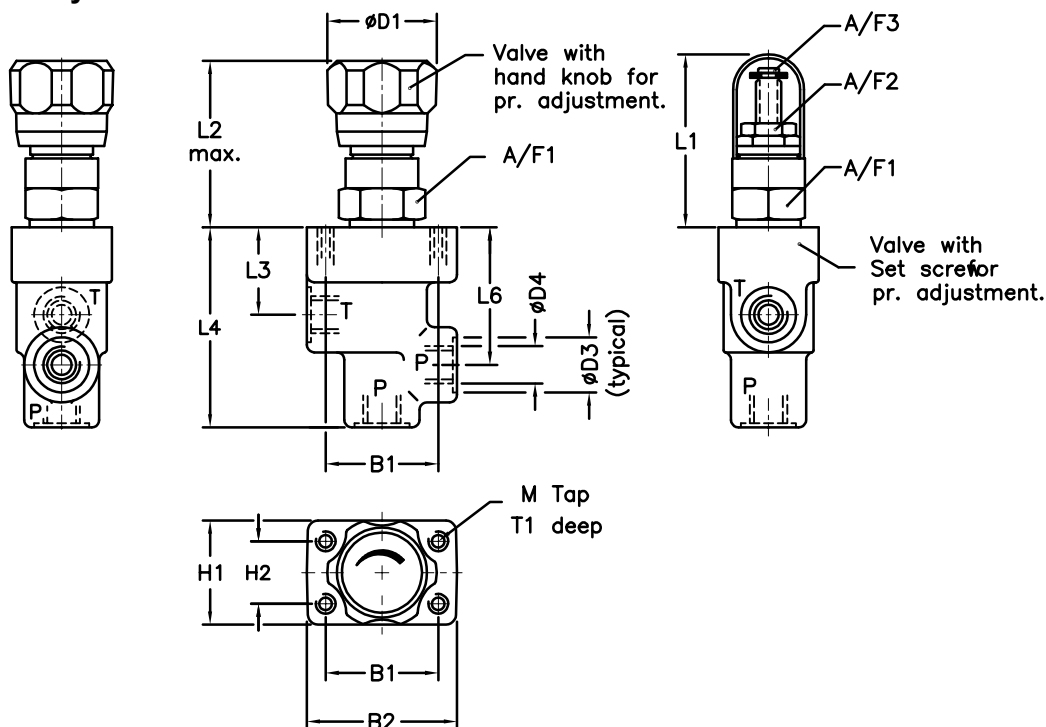
polyhydon pvt. ltd.

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Machhe, Belgaum - 590 014. INDIA.

Phone : +91-(0)831- 2411001
E-mail : sales@polyhydon.com
Website : www.polyhydon.com

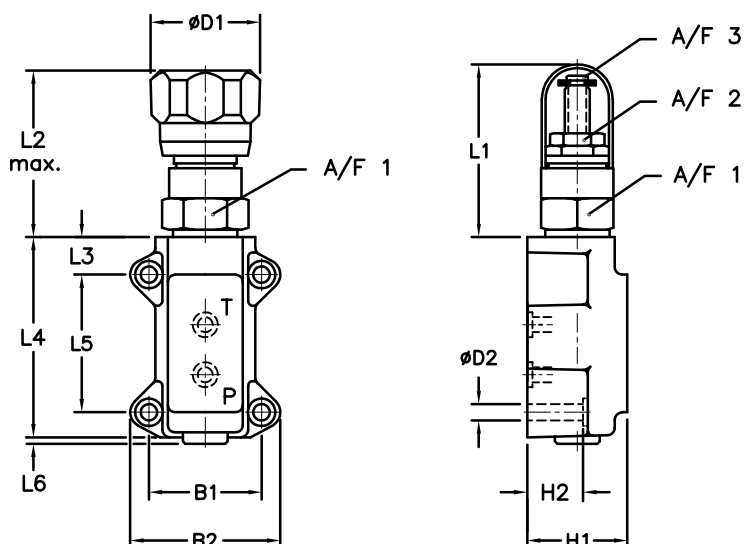


Threaded port body

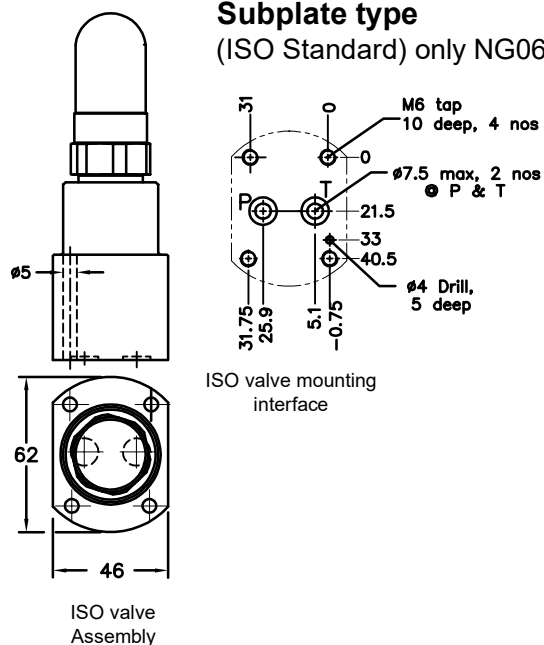


Size	Mass kg	B1	B2	ØD1	ØD2	ØD4	M	A/F1	A/F2	A/F3	L1	L2	L3	L4	L6	H1	H2	T1
06	1.2	45	60	43	22	G1/4	M6	30	19	4	65	64	35	80	55	40	25	11
10	2.8	60	80	48	30	G1/2	M8	36	19	5	80	82	42	100	69	60	40	15
20	6.2	70	95	60	44	G1	M8	46	19	5	78.5	85	54	135	85	70	50	15

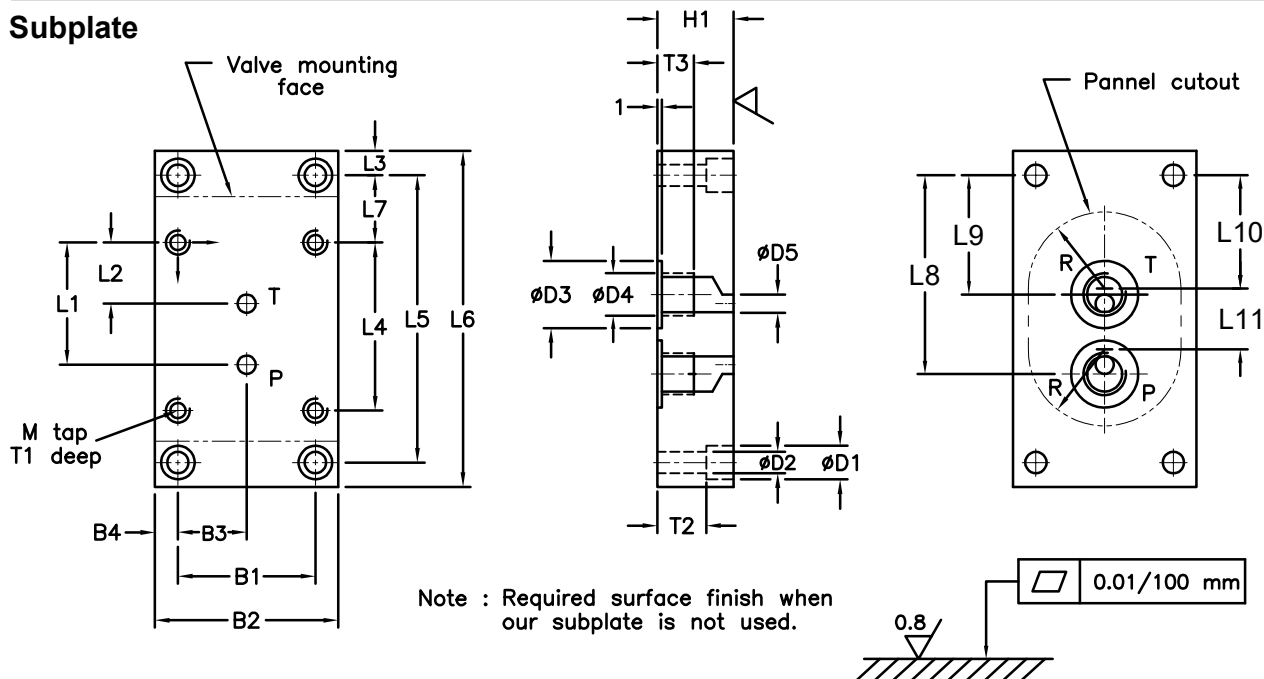
Subplate type (Factory Standard)



Subplate type (ISO Standard) only NG06



Size	Mass kg	B1	B2	ØD1	ØD2	A/F1	A/F2	A/F3	L1	L2	L3	L4	L5	L6	H1	H2	T1
06	1.1	45	60	43	6.6	30	19	4	65	64	15	80	55	04	40	21	11
10	2.4	60	80	48	9	36	19	5	80	82	20	100	70	05	55	28	15
20	8	70	100	60	9	46	19	5	78.5	85	20	135	100	05	70	70	15

**Subplate**

Size	Size	Mass	B1	B2	B3	B4	M	ØD1	ØD2	ØD3	ØD4	ØD5	H1	L1	L2	L3	L4	L5
06	1NSR03G02	1.2	45	60	22.5	7.5	M6	11	7	22	G1/4	06	35	40	20	08	55	94
10	1NSR05G03	2.4	60	80	30	10	M8	11	7	25	G3/8	10	42	45	21	10	70	115
10	1NSR05G04	2.4	60	80	30	10	M8	11	7	30	G1/2	10	42	45	21	10	70	115
20	1NSR07G08	5.3	70	100	35	15	M8	17.5	11.5	44	G1	20	54	65	34	15	100	140

Size	Size	L6	L7	L8	L9	L10	L11	T1	T2	T3	R	Valve mounting screws S.H.C.S.				Tightning Torque		
06	1NSR03G02	1.2	45	60	22.5	7.5	M6	11	7	22	G1/4	06	35	25	20	08	55	94
10	1NSR05G03	2.4	60	80	30	10	M8	11	7	25	G3/8	10	42	30	21	10	70	115
10	1NSR05G04	2.4	60	80	30	10	M8	11	7	30	G1/2	10	42	30	21	10	70	115
20	1NSR07G08	5.3	70	100	35	15	M8	17.5	11.5	44	G1	20	54	40	34	15	100	140

Technical specification

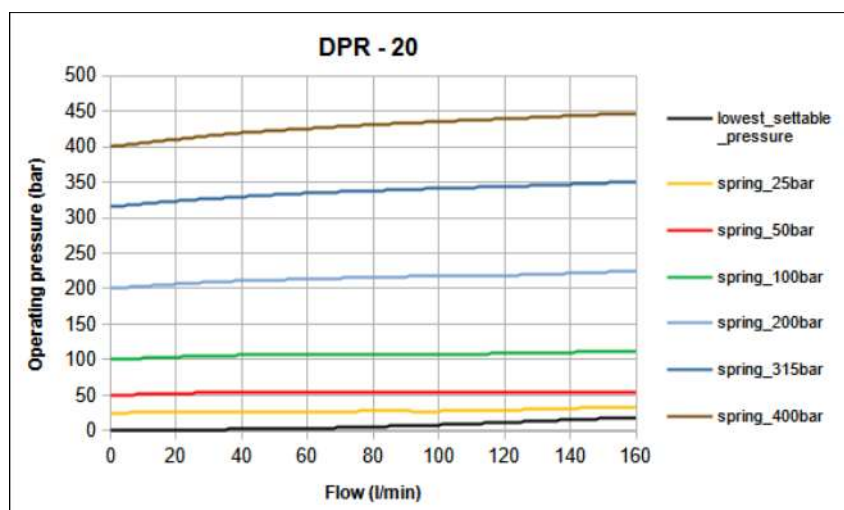
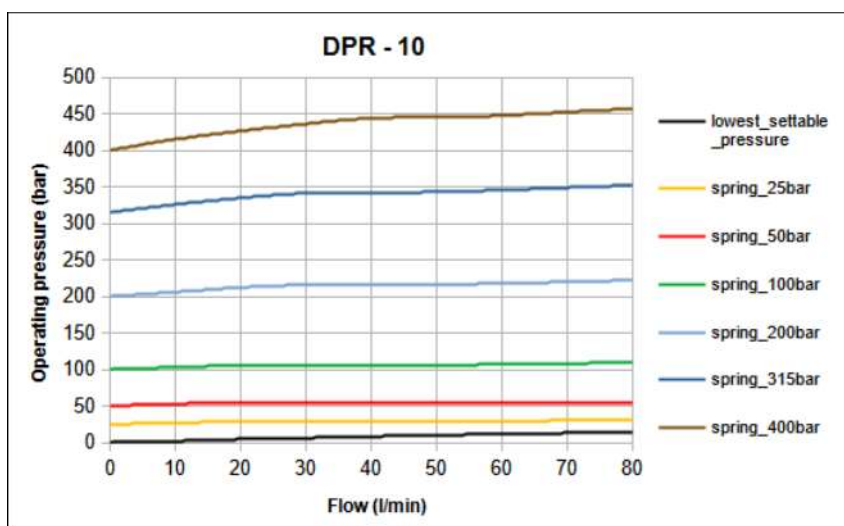
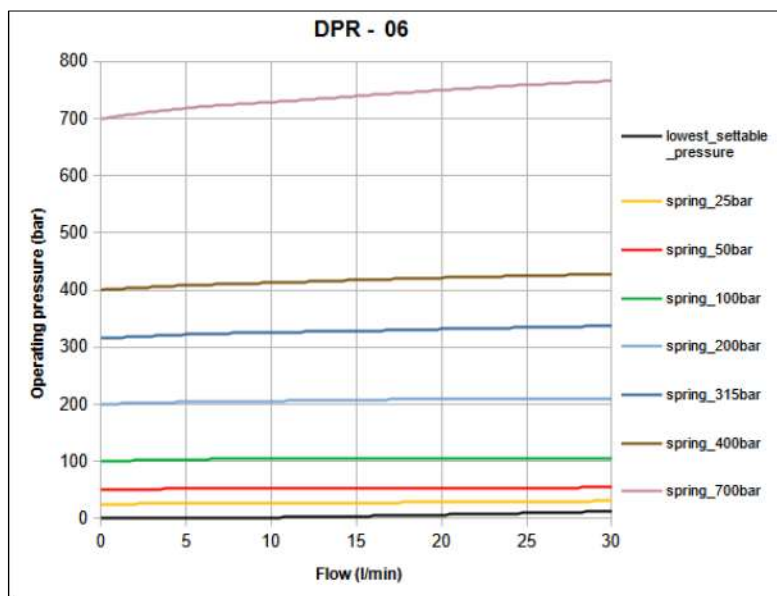
Construction	-	Direct acting, Poppet type.
Mounting style	-	Threaded Cartridge, Threaded Port body or Subplate type (Factory Std. & ISO Std.)
Interface	-	Factory standard
		Note : Subplate mounting valves having interface as per ISO 6264-03-04 and 6264-06-09 are also available on request. Please ask for details
Mounting position	-	Optional.
Flow direction		From P to T
Operating pressure for port		400 bar (Std model)
for port P		700 bar max (for Model DPR06*700-11 only)
for port T		315 bar
Pressure setting range		Upto 25, 50, 100, 200, 315, 400 and 700 bar max.
Hydraulic medium		Mineral Oil
Viscosity range		10 Cst. to 380 Cst
Fluid temperature range		-20°C to + 70°C
Fluid cleanliness requirement		As per ISO 16/13 or better.
Mass		Refer individual table.



Characteristic Curves:

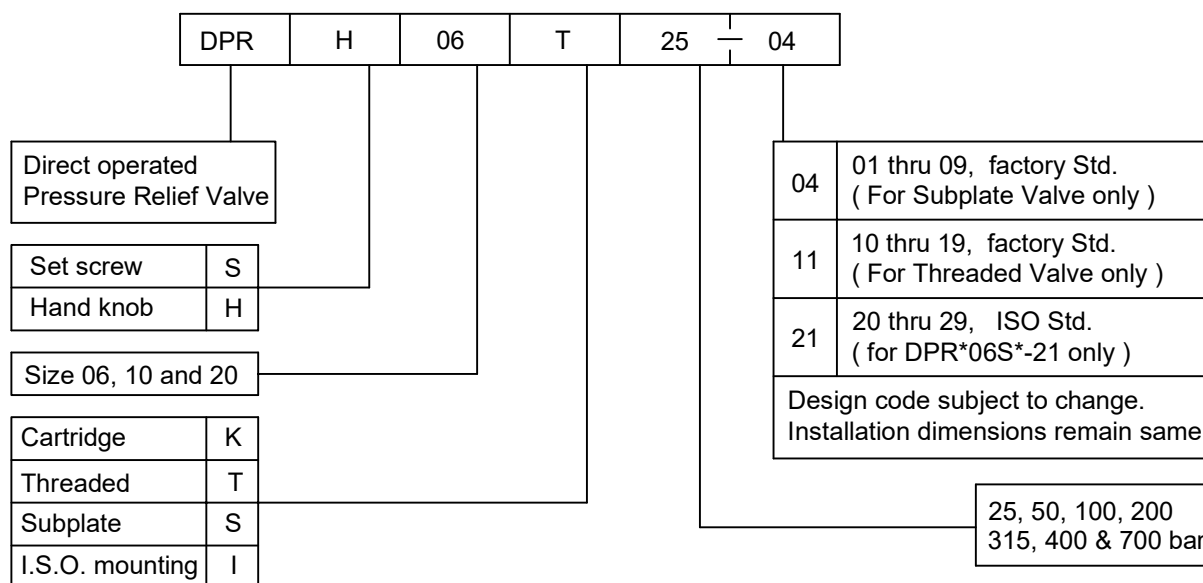
Conditions considered:

- These curves relates only with Cartridge.
- Oil used VG46.
- Temperature ($30 \pm 5^\circ\text{C}$).
- Output pressure Zero.





Ordering code



Note : Subplate to be ordered seperately

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